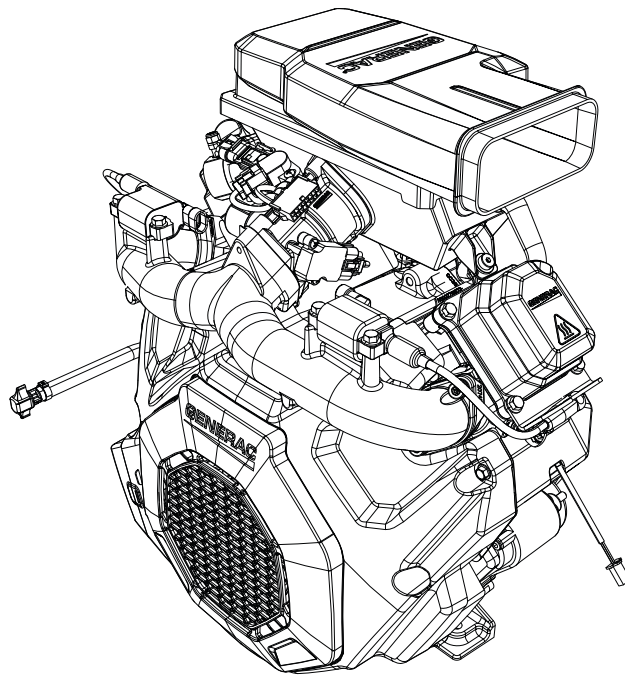


Service Manual

997cc Engine



SAVE THIS MANUAL FOR FUTURE REFERENCE

 CALIFORNIA WARNING

This product can expose you to chemicals including benzene, a carcinogen and reproductive toxicant, which are known to the State of California to cause cancer and birth defects or other reproductive harm.

For more information, go to:
www.p65warnings.ca.gov

(W000808)

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Section 13: Troubleshooting

Section 1: Forward

Introduction

This manual has been written and published by Generac Power Systems, Inc. ("Generac®") to aid dealers, mechanics, and company service personnel when servicing products described herein.

It is assumed these personnel are familiar with servicing procedures for these products, or similar products, manufactured and marketed by Generac. It is also assumed they have been trained in the recommended servicing procedures for these products, which includes the use of mechanics' hand tools and any special tools required.

Correct service and repair is important to the safe, economical, and reliable operation of the products described herein. Troubleshooting, testing, service, and repair procedures recommended by Generac and described in this manual are effective methods of performing such operations. Some of these operations or procedures may require use of specialized equipment. Such equipment should be used when and as recommended.

Generac could not possibly know of and advise the service trade of all conceivable procedures or methods by which a service might be performed, nor of any possible hazards and/or results of each procedure or method. Generac has not undertaken any such wide evaluation. Therefore, anyone who uses a procedure or method not recommended by the manufacturer must first satisfy themselves of neither their safety, nor the safety of those around them or the product's safety will be endangered by the service or operating procedure selected.

Some components or assemblies of the product described in this manual may not be considered repairable. Disassembly, repair, and reassembly of such components may not be included in this manual.

Engines described herein may be used to power a wide variety of products. Service and repair instructions relating to any such products are not covered in this manual. For information pertaining to use of these engines with other products, see owner's or service manuals pertaining to said products.

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

Intended Use

This engine was designed and manufactured for specific applications. Do not attempt to modify the equipment or use it for any application for which it was not designed.

Generac does not approve or authorize use of these engines on All Terrain Vehicles (ATVs), go-carts, motorbikes, aircraft products, personal watercraft, or vehicles intended for use in competitive events. Use of this product in any other than its intended application will void the warranty. Use of these engines in such applications could result in property damage, serious injury (including paralysis), or even death. Do not remove or otherwise alter any safety devices on this engine or any unit it is installed into.

Service Information

Service on this engine within and after the warranty period can be performed by any Independent Authorized Service Dealer (IASD). Service technicians are factory trained and capable of handling all service needs.

Always supply the complete model number and serial number of your unit as given on its data plate decal (A) when contacting an IASD about parts and service.

Warranty statement for this engine is included with the documents provided with the unit.

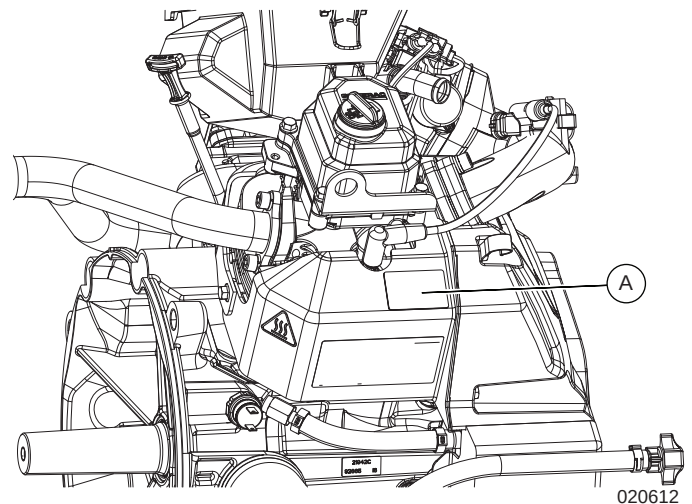


Figure 1-1. Data Plate Decal

Scope

Servicing information provided in this manual applies only to the Generac engine itself. Topics such as air cleaner maintenance, fuel tank maintenance, control panel, and unit rebuild are not included. For servicing information on those components, see diagnostic manual for the associated Generac product.

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Section 2: Safety Rules & General Information

Notice to Users

Study these safety rules carefully before operating or servicing this equipment. Become familiar with the owner's manual and with the engine. The engine can operate safely, efficiently, and reliably only if it is correctly operated and maintained. Many accidents are caused by failing to follow simple and fundamental rules or precautions.

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

Read This Manual Thoroughly



⚠ WARNING

Consult Manual. Read and understand manual completely before using product. Failure to completely understand manual and product could result in death or serious injury. (W000100)

If any portion of this manual is not understood, contact the nearest Independent Authorized Service Dealer (IASD) for starting, operating, and servicing procedures. The owner is responsible for correct maintenance and safe use of the unit.

This manual must be used in conjunction with all other supporting product documentation supplied with the product.

IMPORTANT SAFETY INSTRUCTIONS. SAVE THESE INSTRUCTIONS for future reference. This manual contains important instructions which must be followed during placement, operation, and maintenance of the unit and its components. Always supply this manual to any individual who will use this unit, and instruct them on how to correctly start, operate, and stop the unit in case of emergency.

Emissions

The United States Environmental Protection Agency (US EPA) (and California Air Resources Board (CARB), for engines/equipment certified to California standards) requires this engine/equipment to comply with exhaust and evaporative emissions standards. Locate the emissions compliance decal on the engine to determine applicable standards. See the included emissions warranty for emissions warranty information. Follow the maintenance specifications in this manual to verify the

engine complies with applicable emissions standards for the duration of the product's life.

Safety Rules

The manufacturer cannot anticipate every possible circumstance that might involve a hazard. The alerts in this manual, and on tags and decals affixed to the unit, are not all inclusive. If using a procedure, work method, or operating technique that the manufacturer does not specifically recommend, verify that it is safe for others and does not render the equipment unsafe.

Throughout this publication, and on tags and decals affixed to the unit, DANGER, WARNING, CAUTION, and NOTE blocks are used to alert personnel to special instructions about a particular operation that may be hazardous if performed incorrectly or carelessly. Observe them carefully. Alert definitions are as follows:

⚠ DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury. (D000001)

⚠ WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury. (W000002)

⚠ CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury. (C000003)

NOTE: Notes contain additional information important to a procedure and will be found within the regular text of this manual.

These safety alerts cannot eliminate the hazards that they indicate. Common sense and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

Non-Generac Parts and Equipment

Generac products obtain listings and certifications through an evaluation of the original design and configuration of the product. Any modification to this product, including the incorporation of non-Generac parts or equipment, may void this product's listing or certification. Damage or failures caused by any

modification to the product are not covered by the product's warranty. See limited warranty for more information.

General Hazards



Automatic start-up. Disconnect utility power and render unit inoperable before working on unit. Failure to do so will result in death or serious injury.
(D000191)



Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.
(D000103)



Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.
(D000129)



Accidental Start-up. Disconnect the negative battery cable, then the positive battery cable when working on unit. Failure to do so could result in death or serious injury.
(W000130)



Equipment damage. Only qualified service personnel may install, operate, and maintain this equipment. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage.
(W000182)



Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.
(W000111)



Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury.
(W000115)



Equipment and property damage. Do not alter construction of, installation, or block ventilation for generator. Failure to do so could result in unsafe operation or damage to the generator.
(W000146)



Risk of injury. Do not operate or service this machine if not fully alert. Fatigue can impair the ability to service this equipment and could result in death or serious injury.
(W000215)



Accidental Start-Up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.
(W000141)

IMPORTANT NOTE: Incorrect treatment of this engine can damage and shorten its life.

Contact an IASD with questions about intended use.

Replacement Parts

This equipment is designed to be used with Generac authorized parts ONLY. User assumes all risks and liabilities if equipment is used with parts which DO NOT comply with minimum specifications. During any maintenance procedure, replacement fasteners must have the same measurements and strength as the removed fasteners. Metric bolts and nuts have numbers indicating their strength. Imperial bolts use radial lines to indicate strength, while most Imperial nuts do not have strength markings. Mismatched or incorrect fasteners can cause damage, malfunction, and possible injury.

Exhaust Hazards



Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.
(D000103)


⚠ DANGER

Asphyxiation. Carbon monoxide can kill in minutes. Operate this unit outdoors only. Failure to do so will result in death or serious injury.

(D000525)


⚠ WARNING

Asphyxiation. Always use a battery operated carbon monoxide alarm indoors and installed according to the manufacturer's instructions. Failure to do so could result in death or serious injury.

(W000178)


⚠ WARNING

Asphyxiation. Do not inhale vapor or fumes. Seek medical attention immediately. Failure to do so could result in death or serious injury.

(W000342)

⚠ CAUTION

Personal injury. Wear appropriate personal protective equipment at all times while operating and servicing unit. Failure to do so could result in personal injury.

(C000419)

- Carbon monoxide produced during use can kill in minutes. NEVER use indoors, in a vehicle, or in other sheltered areas, EVEN IF doors and windows are open. Only use OUTSIDE and far away from windows, doors, and vents.
- Breathing carbon monoxide can cause headache, fatigue, dizziness, vomiting, confusion, seizures, nausea, fainting, or death. If you start experiencing any of these symptoms, IMMEDIATELY move to fresh air. Seek medical attention.
- Keep exhaust gas from entering a confined area through windows, doors, ventilation intakes, or other openings. Warn any occupants inside about the symptoms of carbon monoxide so they know to move to fresh air if they begin to feel ill.
- Use a respirator or mask whenever there is a possibility of inhaling harmful gas or vapors.
- Read all instructions with respirator or mask to verify respirator or mask will provide the necessary protection against inhaling harmful gas or vapors. IMMEDIATELY move to fresh air if experiencing any of the CO symptoms listed above.

Electrical Hazards


⚠ DANGER

Electrocution. Water contact with a power source, if not avoided, will result in death or serious injury.

(D000104)

- Never service this engine or handle any electrical device while standing in water, while barefoot, or while hands or feet are wet. DANGEROUS ELECTRIC SHOCK will result.
- Never operate the engine (a) in the rain; (b) in any enclosed compartment; (c) if the engine speed changes; (d) if the engine sparks; (e) if flame or smoke is observed while the engine is running.

Fire and Fuel Hazards


⚠ DANGER

Explosion and fire. Fuel and vapors are extremely flammable and explosive. Add fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.

(D000105)


⚠ DANGER

Explosion and fire. Fuel and vapors are extremely flammable and explosive. No leakage of fuel is permitted. Keep fire and spark away. Failure to do so will result in death or serious injury.

(D000192)


⚠ DANGER

Explosion and Fire. Fuel and vapors are extremely flammable and explosive. Store fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.

(D000143)


⚠ DANGER

Explosion and Fire. Do not fill fuel tank past full line. Allow for fuel expansion. Overfilling may cause fuel to spill onto engine causing fire or explosion, which will result in death or serious injury.

(D000214)



⚠ DANGER

Risk of fire. Allow fuel spills to completely dry before starting engine. Failure to do so will result in death or serious injury.

(D000174)



⚠ DANGER

Explosion and Fire. Drain fuel in a well ventilated area, into an approved container only after engine has cooled. Failure to do so will result in death or serious injury.

(D000180)



⚠ WARNING

Explosion and Fire. Do not smoke while refueling unit. Failure to do so could result in death, serious injury, or property or equipment damage.

(W000284)



⚠ WARNING

Fire risk. Fuel and vapors are extremely flammable. Do not operate indoors. Doing so could result in death, serious injury, or property or equipment damage.

(W000281)

⚠ CAUTION

Engine Damage. Do not operate with incorrect fuel. Doing so will result in reduced service life of the engine.

(C000569)

- Avoid spilling gasoline on a hot engine. Comply with all of the laws regulating storage and handling of gasoline.
- It is a violation of California Public Resource Code, Section 4442, to use or operate the engine on any forest-covered, brush-covered, or grass-covered land unless the exhaust system is equipped with a spark arrester, as defined in Section 4442, maintained in effective working order. Other states or federal jurisdictions may have similar laws. Contact the original equipment manufacturer, retailer, or an IASD to obtain a spark arrester designed for the exhaust system installed on this engine.

Hearing Loss Hazard



⚠ WARNING

Hearing Loss. Hearing protection is recommended when using this machine. Failure to wear hearing protection could result in permanent hearing loss.

(W000107)

Moving Parts Hazard



⚠ WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.

(W000111)



⚠ WARNING

Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury.

(W000115)

Before Starting Equipment

⚠ WARNING

Equipment damage. Do not attempt to start or operate a unit in need of repair or scheduled maintenance. Doing so could result in serious injury, death, or equipment failure or damage.

(W000291)

⚠ WARNING

Personal Injury. Never rotate engine with spark plug removed. Rotating engine with spark plug removed could result in death or serious injury.

(W000564)

⚠ CAUTION

Engine damage. Verify proper type and quantity of engine oil prior to starting engine. Failure to do so could result in engine damage.

(C000135)

- Verify spark plug, muffler, fuel cap, and air cleaner are in place.

When Operating Equipment

WARNING

Equipment damage. Operating engine with obstructed air cooling slots could result in death, serious injury, and unit damage.

(W000255)



WARNING

Fire hazard. Do not obstruct cooling and ventilating airflow around the engine. Inadequate ventilation could result in fire hazard, possible equipment damage, death or serious injury.

(W000256)

WARNING

Personal injury and equipment damage. Do not tamper with engine governed speed. Operating engine too fast or slow increases the risk of injury or permanent engine damage.

(W000254)

CAUTION

Equipment damage. Never stop the engine by moving throttle control lever to CHOKE. Doing so could result in engine damage.

(C000457)

- DO NOT tip engine or equipment at angle which causes fuel to spill.
- Reflective exhaust heat may damage the fuel tank, causing fire. Keep at least 5 ft (1.3 m) of clearance on all sides of the engine for adequate cooling, maintenance, and servicing.

Hot Surfaces Hazard



WARNING

Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire.

(W000108)

- AVOID hot exhaust gases.
- Allow equipment to cool before touching.

When Transporting or Repairing Equipment

WARNING

Accidental Start-Up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.

(W000141)

- Transport/repair with fuel tank EMPTY.

Spark Hazards

WARNING

Accidental Start-Up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.

(W000141)

When Testing For Engine Spark



WARNING

Explosion. Turn fuel supply OFF before checking for spark. Failure to do so could result in death or severe injury.

(W000333)

WARNING

Electric Shock. Never test ignition spark with the spark plug or spark plug wire(s) removed. Always use an approved spark tester. Failure to do so could result in death, serious injury, or equipment damage.

(W000406)

- Use approved spark plug tester.
- Place the controller in MANUAL mode.

Battery Hazards



WARNING

Explosion. Do not dispose of batteries in a fire. Batteries are explosive. Electrolyte solution can cause burns and blindness. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(W000162)



WARNING

Explosion. Batteries emit explosive gases while charging. Keep fire and spark away. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(W000137)



⚠ WARNING

Electrical shock. Disconnect battery ground terminal before working on battery or battery wires. Failure to do so could result in death or serious injury.

(W000164)



⚠ WARNING

Risk of burns. Batteries contain sulfuric acid and can cause severe chemical burns. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(W000138)

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit Call2Recycle website at: <http://Call2Recycle.org/locator>.

General Maintenance

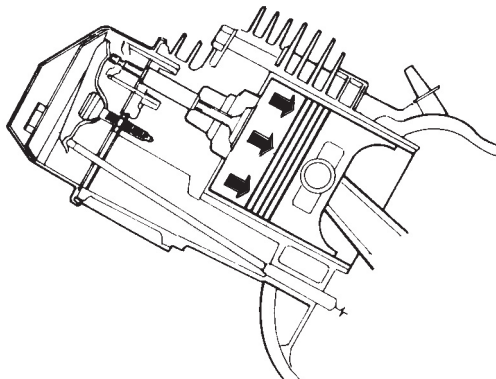
- Inspect engine periodically. Repair or replace all damaged or faulty parts immediately.
- Inspect fuel system frequently for leaks or damage. Repair or replace any damaged or leaking component immediately. Never attempt to change, alter, or modify the engine fuel system in any way which might affect safety or compliance with applicable codes and standards.

Section 3: Four-Cycle Engine Theory

Introduction

Four events must occur in the correct sequence and at the correct time for the engine to run correctly. These events are (A) intake, (B) compression, (C) power, and (D) exhaust.

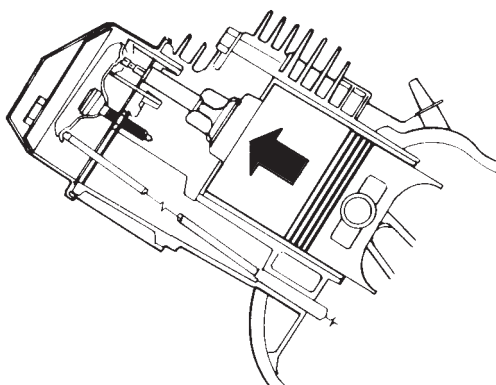
(A) Intake



001684

The piston is traveling from top dead center (TDC) to bottom dead center (BDC). The cam has opened the intake valve. The piston's downward movement in the cylinder creates a partial vacuum in the cylinder. Air at atmospheric pressure is drawn into the cylinder through the airbox and is mixed with fuel, supplied by the injector, as it enters the cylinder. The intake stroke is complete when the piston reaches BDC.

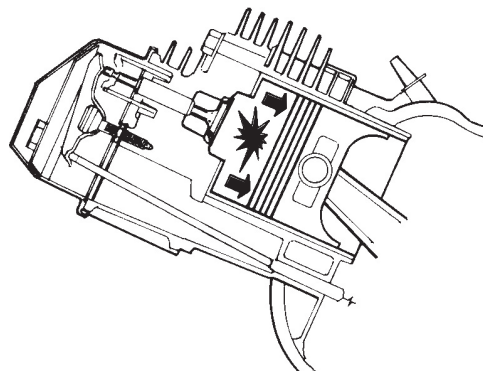
(B) Compression



001685

As the piston reaches BDC, both the intake and exhaust valves are closed. The piston moves upward toward TDC and the fuel-air mixture is compressed. Ignition occurs just before the piston reaches TDC.

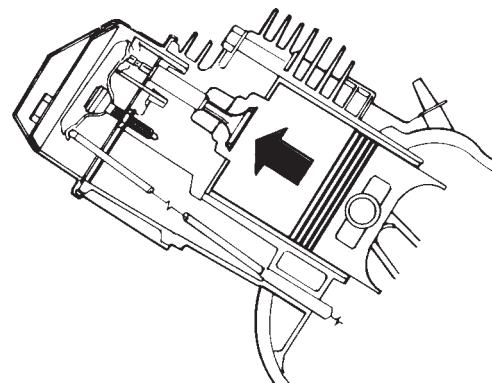
(C) Power



001686

Combustion is already in progress when the piston reaches TDC. The intake and exhaust valves remain closed as expanding gases of combustion force the piston downward.

(D) Exhaust



001687

The exhaust stroke begins when the piston has reached BDC and has started its upward movement. The intake valve is closed, and the exhaust valve is open to let gases escape.

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Section 4: General Information

Maintenance Recommendations

See the appropriate unit owner's manual for break-in and maintenance intervals, and oil and fuel recommendations.

Verify generator is in SERVICE mode via the Field Pro app or the control panel SERVICE button prior to starting any preventative or corrective maintenance.

Checking Engine Oil



Skin Irritation. Avoid prolonged or repeated contact with used motor oil. Used motor oil has been shown to cause skin cancer in laboratory animals. Thoroughly wash exposed areas with soap and water.

(W000210)

IMPORTANT NOTE: Do not overfill. Excessive oil will foam, smoke, and eventually damage internal engine parts.

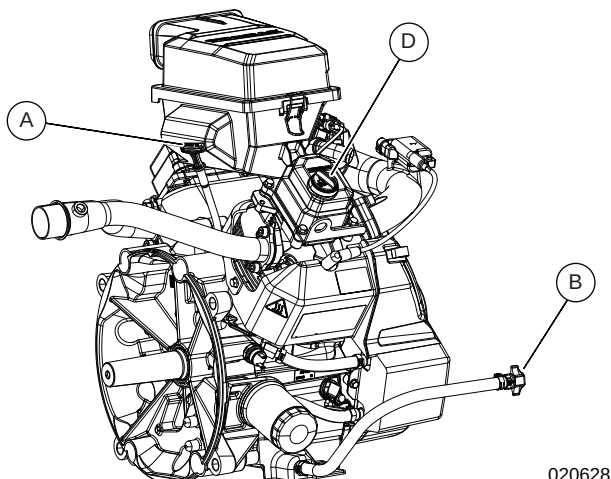
See [Specifications](#) for crankcase oil capacities. Do not use special additives. Verify unit is level when filling oil.

Change oil and filter after first 25 hours of operation for break-in, then resume normal maintenance interval. Change oil and filter more frequently if engine is operated in dirty or dusty conditions, operated under heavy loads, or in high ambient air temperatures.

Inspecting Engine Oil Level

Proceed as follows to inspect engine oil level prior to each use, or every eight (8) hours of operation:

1. See [Figure 4-1](#). Clean area around engine dipstick (A) and oil fill cap (D).

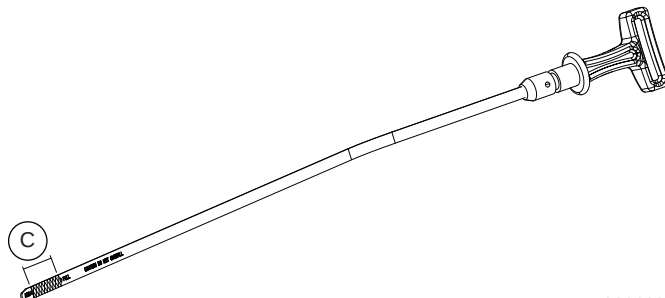


020628

Figure 4-1. Oil Dipstick and Oil Drain Cap

2. Remove oil dipstick (A) and wipe oil dipstick clean.

3. See [Figure 4-2](#). Replace oil dipstick. Remove oil dipstick and verify oil level is within safe operating range (C).



020629

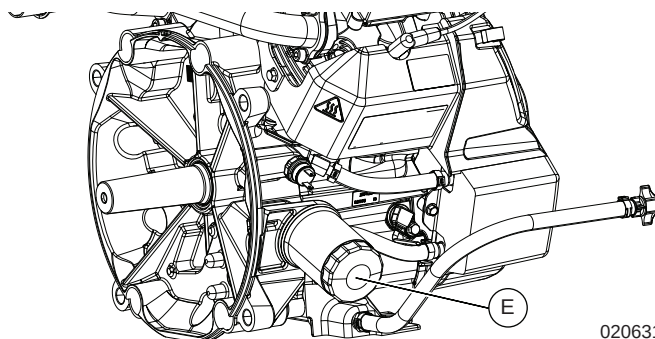
Figure 4-2. Safe Operating Range

4. See [Figure 4-1](#). Add recommended engine oil as necessary in fill location (D).
5. Install oil dipstick.

Changing Oil and Oil Filter

Proceed as follows to change oil and oil filter:

1. See [Figure 4-1](#). Remove oil drain cap (B) and drain oil while engine is still warm.
2. See [Figure 4-3](#). Remove used oil filter (E) and discard per local, state, or national regulations.



020631

Figure 4-3. Oil Filter

3. See [Figure 4-1](#). Install oil drain cap (B).
4. Lightly lubricate new oil filter gasket with fresh, clean engine oil before installing new oil filter.
5. Screw new oil filter on by hand until oil filter gasket contacts oil filter adapter. Tighten three-quarters to one full turn farther.
6. See [Figure 4-1](#). Remove oil fill cap (D) and slowly fill with new oil, of required amount, of required service classification and viscosity grade.
7. Start and run engine at idle for 30 seconds. Stop engine. Verify oil level and add if required.
8. Start engine and inspect for oil leaks.

Inspecting and Cleaning Intake Screen

See [Figure 4-4](#). Continued operation with a clogged cooling system can cause severe overheating and possible engine damage. Inspect and clean the intake screen (A) during routine and periodic maintenance.

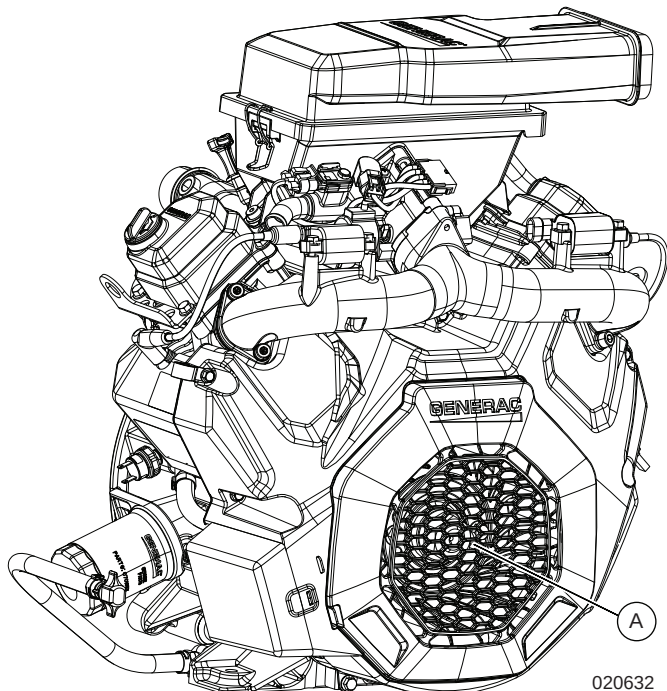


Figure 4-4. Intake Screen and Oil Cooler Fins

Replacing Spark Plugs

Replace spark plug if electrode is burned away or if porcelain is cracked. See generator owner's manual for spark plug replacement specification.

See [Figure 4-5](#). Inspect spark plug gap and set spark plug gap to 0.020 in (0.50 mm) for all models. Tighten spark plug to 168 in-lb (18.9 Nm).

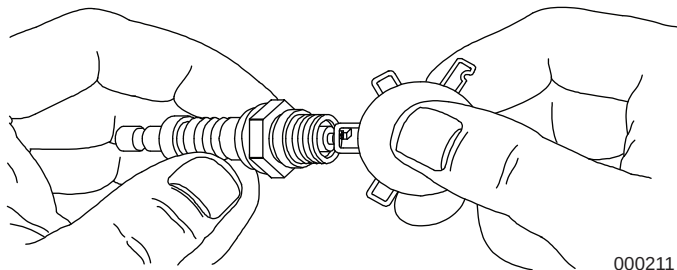


Figure 4-5. Inspecting Spark Plug Gap

Maintaining Air Cleaner



⚠ WARNING

Risk of fire. Never operate engine without the air cleaner installed. Operating engine without the air cleaner could result in death or serious injury.

(W000249)

A correctly serviced air cleaner protects internal engine parts from dirt and dust particles in the air. If air cleaner instructions are not carefully followed, dirt and dust normally collected in the air cleaner will be drawn into the engine. Particulates will wear on cylinder bores and piston rings, allowing particles to enter the crankcase and contaminate the oil, forming an abrasive mixture which will quickly wear internal moving parts.

Inspect the air cleaner on every engine serviced or repaired. If the air cleaner shows signs of neglect, show it to the user before cleaning. Instruct the user on correct care techniques to assure long engine life.

NOTE: Replace worn or damaged air cleaner gaskets and mounting gaskets to prevent dirt and dust from entering the engine due to incorrect sealing. Replace bent air cleaner mounting bracket if necessary.

Troubleshooting

Most complaints concerning engine operation can be classified as one or a combination of the following:

- Will not start
- Hard starting
- Lack of power
- Runs rough
- Vibration
- Overheating
- Noise
- High oil consumption

NOTE: What appears to be an engine malfunction may be a fault of the equipment connected to the engine rather than the engine. If equipment is suspect, see equipment affecting engine operation.

Systematic Inspection

If engine will not start and the cause of malfunction is not readily apparent, perform a systematic inspection in the following order:

1. Fuel
2. Ignition
3. Compression

This inspection, performed in a systematic manner, can usually be done in a matter of minutes. It is the quickest

and surest method of determining the cause of failure. The basic inspection procedure is the same for all engine models. Any variation, by model, will be described under the subject heading.

Checking Fuel Pressure

- NG: 3.5 – 7.0 in H₂O (0.87 – 1.74 kPa)
- LP: 10 – 12 in H₂O (2.49 – 2.99 kPa)

For gaseous units, check the following:

- Is there sufficient fuel supply from the meter?
- Is the fuel supply pipe the correct size?
- Are there any bends or kinks in the flexible supply line?

Checking Ignition

See Section 7: Ignition for testing procedures.

Checking Compression

Proceed as follows to check engine compression:

1. Disconnect ECM.
2. Manually open throttle plate on the ETC to wide open throttle (WOT) position.
3. Connect a remote button starter switch from the battery positive to the starter relay.
4. Remove both spark plugs and insert a compression gauge (Generac P/N GN04001) into either cylinder (one cylinder at a time).
5. Turn engine over with engine starter until there is no further increase in pressure. Record this reading.
6. Repeat procedure on other cylinder and record reading.

The compression difference between both cylinders should not exceed 25%. More than 25% indicates loss of compression in the cylinder with lower pressure.

To calculate the difference in percentage, divide the difference in PSI (kPa) by the higher pressure reading. See examples below.

	Cylinder 1	Cylinder 2	Difference
Engine #1	165 psi (1,137.6 kPa)	160 psi (1,103.1 kPa)	5 psi (34.5 kPa)

Engine #2	175 psi (1,206.6 kPa)	155 psi (1,068.7 kPa)	20 psi (137.9 kPa)
Engine #3	160 psi (1,103.1 kPa)	112 psi (772.2 kPa)	48 psi (330.9 kPa)

Engine #1: (5 psi ÷ 165 psi) [34.5 kPa ÷ 1,103.1 kPa] = 0.03 = 3% difference.

Engine #2: (20 psi ÷ 175 psi) [137.9 kPa ÷ 1,206.6 kPa] = 0.11 = 11% difference.

Engine #3: (48 psi ÷ 160 psi) [330.9 kPa ÷ 1,103.1 kPa] = 0.30 = 30% difference

Using this method of calculation, Cylinder 2 in Engine #3 has lost significant compression.

If compression is poor, look for one or more of the following causes:

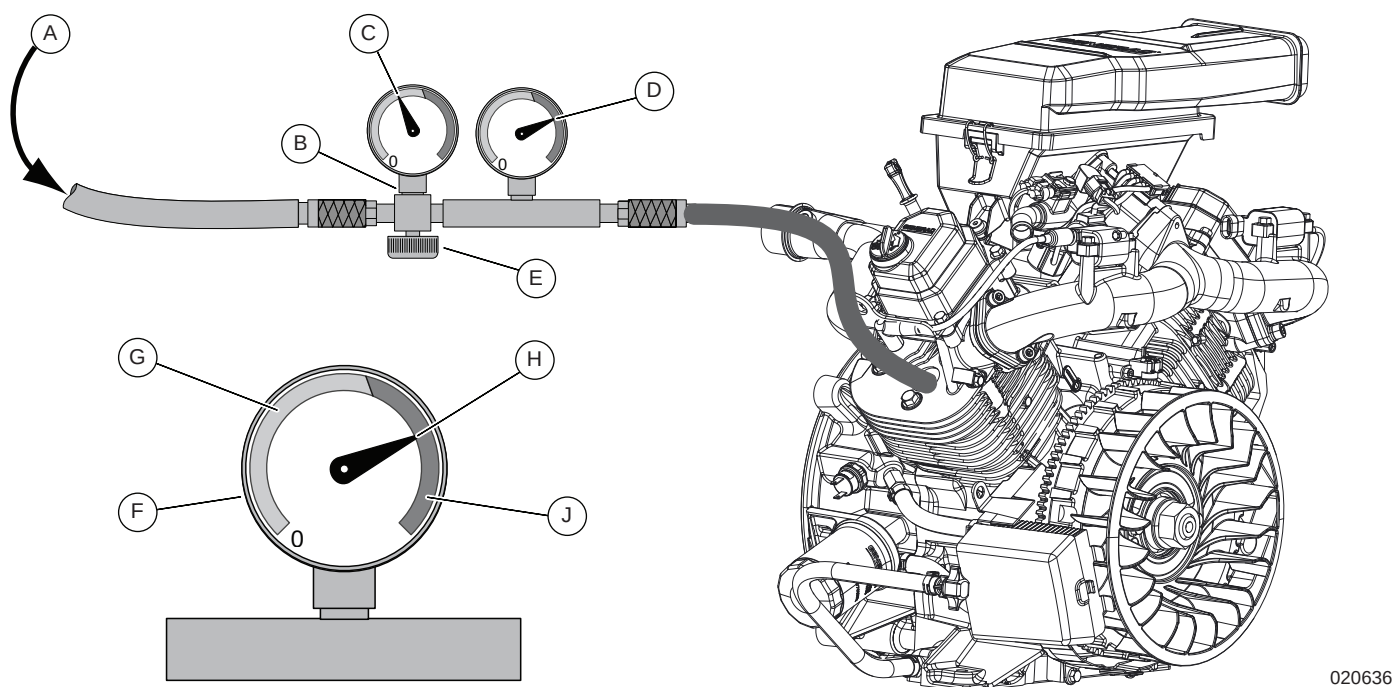
- Loose cylinder head bolts
- Failed cylinder head gasket
- Burned / worn valves or valve seats
- Insufficient valve clearance
- Warped cylinder head
- Warped valve stem or worn valve guides
- Worn or broken piston ring(s)
- Worn or damaged cylinder bore
- Broken connecting rod

Model	Normal Compression
997 HSB	180 – 220 psi (1,240 – 1,520 kPa)

Cylinder Leakdown Test

A cylinder leakdown tester (Generac P/N 0F77000SRV) may be used to test the sealing capability of compression components of each cylinder while at TDC of the compression stroke, and quickly identify the problem component. A cylinder leakdown test will assist in troubleshooting leaking valves or rings.

See [Figure 4-6](#). There are two gauges on the tester. One measures incoming air pressure, and the other measures the percent of cylinder leakage.



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Figure 4-6. Cylinder Leakdown Tester

A	Compressed air inlet
B	Air pressure regulator
C	Inlet gauge pressure set point
D	Outlet gauge pressure
E	Regular adjustment knob
F	Outlet gauge
G	Red range indicates unacceptable air leakage
H	Needle indicates minimal air leakage
J	Green range indicates acceptable air leakage
NOTE: Remove blower housing and cylinder wrappers before proceeding with test. See Section 6: Cylinder Wrappers and Blower Housing.	

Testing Procedure



⚠ WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.

(W000111)

Proceed as follows to perform a cylinder leakdown test:

1. Disconnect the ECM.
2. Manually open the throttle plate on the ETC.
3. Connect a remote button starter switch from the battery positive (+) to the starter relay.
4. Align piston at TDC.
5. Install cylinder leakdown tester in the spark plug hole.
6. Pressurize cylinder to 90 psi (621 kPa).
7. Observe cylinder leakage and identify location of air leak.

NOTE: If leaking into intake or exhaust port, check valve face and valve seat condition.

Results

- 0–10%—Cylinder compression is acceptable
- 10–30%—Excessive cylinder leakage may exist
- 30–100%—Cylinder requires repair

Factors Affecting Cylinder Condition

- EFI
- Crankcase vacuum
- Ignition timing
- A partially sheared flywheel key will affect ignition timing and engine performance.
- Spark plug
 - A fouled spark plug may indicate fuel injection issues or engine cylinder concerns.
- Leak in spark plug wire
- Head gasket
- Intake manifold
- Valves
- Rings
- Piston
- Cylinder Bore

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Section 5: Specifications

Specifications

Model	G-Force TM 1000 Series
Bore	90 mm
Stroke	78.4 mm
Displacement	997 cc
Oil capacity with oil filter change	Approx. 2.2 qt (2.1 L)
Oil capacity without oil filter change	2.3 qt (2.16 L)
Common Specifications	
Crank position sensor gap (not adjustable)	0.007 – 0.06 in (0.19 – 1.42 mm)
Stator resistance	0.4 Ohms
Crankshaft end play	0.009 – 0.022 in (0.24 – 0.56 mm)
Spark plug gap	0.020 in (0.508 mm)
Normal compression	180 – 220 psi (1,240 – 1,520 kPa)
Valve lash	Hydraulically adjusted, no manual setting required.

Torque Specifications

P/N	Name	Location Description	Final Torque	Unit
Longblock				
0L1976	Screw HHFC M8-A.25 X 38LG	Connecting Rod Cap	21.9-26.8	N-m
0C3006	Screw HHFC M10-1.5 x 55 G8.8	Gear Cover	43.9-53.7	N-m
0G0926	Screw HHFCS M6-1.0 x 16 G8.8	Oil Float Sensor / Shield Deflector	6.5-7.5	N-m
0G0919	Screw HHFC M6-1.0 x 12 G8.8	Pressure Relief Ball Cover / Oil Tube Clamp	7.99-10.02	N-m
0D6147	Screw HHFCS M6-1.0 x 10	Dipstick Tube	11-13	N-m
Oil System				
0G6431	Screw SHC M6-1.0 x 50 G8.8	Oil Pump	10-11	N-m
0F5458	Screw Hi-Low #10 x 3/8	Oil Pick-Up	7-11	in-lbs
0L2917D	LOW OIL PRESS SWITCH 10 PSI NO	Oil Pressure Switch	*	*
070185E	OIL FILTER 90	Oil Filter	9.5	N-m
0L6347	3/8-18 NPT	Oil Drain Hose Barb	*	*
0L6346	Barb Str SAE ¼ NPT x 3/8 EXT	Upper Oil Cooler Barb	*	*
0L6345	Barb Str SAE ¼ NPT x 3/8 ST-Z	Lower Oil Cooler Barb	*	*

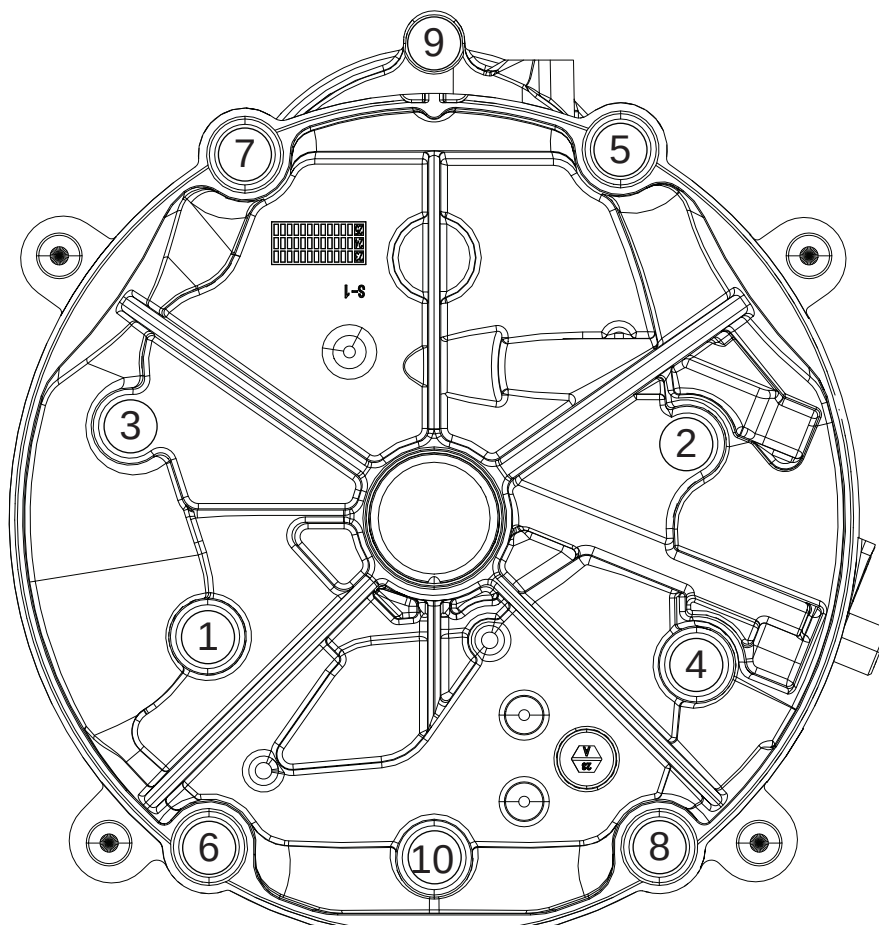
P/N	Name	Location Description	Final Torque	Unit
Cylinder Head				
A0003402218	M8-1.25-70 SEMS HEAD BOLT	Cylinder Head Bolts	26.5-30.4	N-m
A0001336196	Stud Rocker Ball	Rocker Arm Studs	15.1-16.8	N-m
G080318	Screw HHFC M6-1.0 x 25 FTH G8	Rocker Cover	6.5-7.5	N-m
A0003637864	Spark Plug Generac Gap 0.020"	Cylinder Head Spark Plugs	20-30	N-m
0J4967	SCREW HHFC M5-0.8 X 12	Breather Plate	4-4.5	N-m
0D6147	Screw HHFC M6-1.0 x 10	Cylinder Head Temp Sensor	6.5-7.5	N-m
Flywheel				
0L6110	Screw SHC M6-1.0 x 20	Stator Screws	11-13.6	N-m
0E9507	Screw HHFC M8-1.25 X 14	Stator Ground Screw To Crankcase	25-27.5	N-m
A0000283026	Screw SHC M5-0.8 x 12	Ckp Sensor & Wire Retention Bracket	6-6.5	N-m
0H4957	Screw HHFCS M8-1.25 x 10	Flywheel Fan	22-27	N-m
0C3034	Nut Hex M24-2.0 G8	Flywheel Nut	142-175	N-m
G061906	Screw HHC M8-1.25 x 85 C8.8	Starter Motor	25.5-30	N-m
G080318	Screw HHFC M6-1.0 x 25	Ignition Coil	10-12	N-m
Intake System				
0G3730A	Screw SHC M8-1.25 x 30 G8.8 SEMS	Intake Manifold	19-22	N-m
Trim				
0G0926	Screw HHFCS M6-1.0 x 16 G8.8	Backplate To Crankcase / Engine Wrappers	7.99-10.02	N-m
0C9764	Screw HH Hi-Lo 1/4-15 x 3/4	Wrapper To Blower Housing	2.83	N-m
10000034692	Nut Hex Flange M8-1.25 Nyloc	Eng Foot / Neg Batt Cable	24.4-31.2	N-m
0C8563	Screw HHFC M6-1.0 x 12	Wrappers To Cylinder Head	7.99-10.02	N-m

***NPT Fittings:** As a general rule, pipe fittings with tapered threads should not be assembled to a specific torque. Instead, they should be installed by the number of turns past finger tight. All NPT plugs below 1 in size should be installed to 2-3 turns from finger tight (T.F.F.T.).

Torque Sequences

Proceed with the correct sequence to tighten the gear cover:

(1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 — 9 — 10)

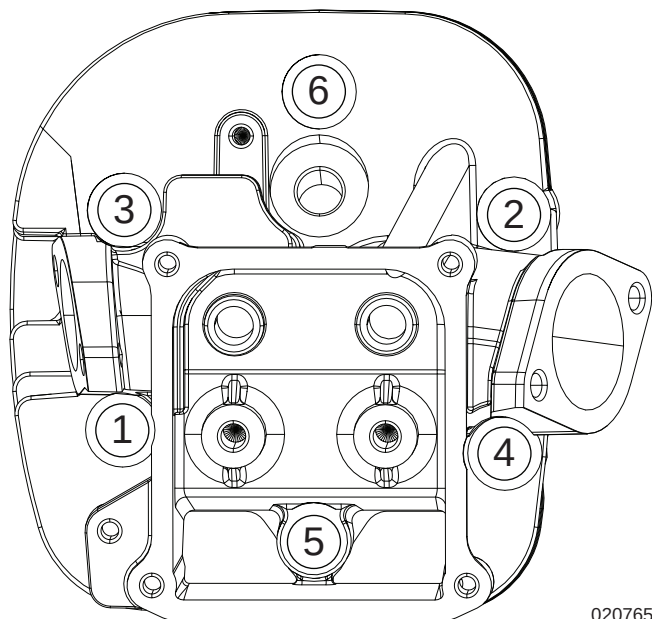


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Figure 5-1. Gear Cover Torque Sequence

Proceed with the correct sequence to tighten the cylinder head:

(1 — 2 — 3 — 4 — 5 — 6)



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Figure 5-2. Cylinder Head Torque Sequence

Component Weights

Part Description	Part Number	Weight (kg)[lbs]
Cylinder Head #1	A0001773455	1.78 [3.9]
Cylinder Head #2	A0002106859	1.82 [4.0]
Complete Engine Assembly (Dry)	A0000683019	44.18 [97.4]

Approved Sealants/Lubricants

- Thread sealants (used for threads which require sealant):
 - Permabond LH150 (or equivalent)
 - Permatex® High Temperature Thread Sealant (or equivalent)
 - LOCTITE® 567™ (or equivalent)
- RTV sealants (used for gasket between crankcase and gear cover):
 - Permatex® Ultra Black® Gasket Maker (or equivalent)
 - Loctite SI 593™ Black (or equivalent)
- Assembly lube (used when applying parts which require lubricant for installation):
 - Lube Tech T24872 (or equivalent)

Section 6: Engine Disassembly

Long Block Disassembly

Proceed as follows to disassemble the long block:

1. Drain oil, disconnect wire harness, unplug connector cables, and disconnect fuel hose.
2. Disconnect intake and exhaust systems from equipment.
3. Remove engine from equipment.

Removing Ignition Coils

Proceed as follows to remove ignition coils:

1. See [Figure 6-1](#). Remove spark plug wires (A) from spark plugs (B).

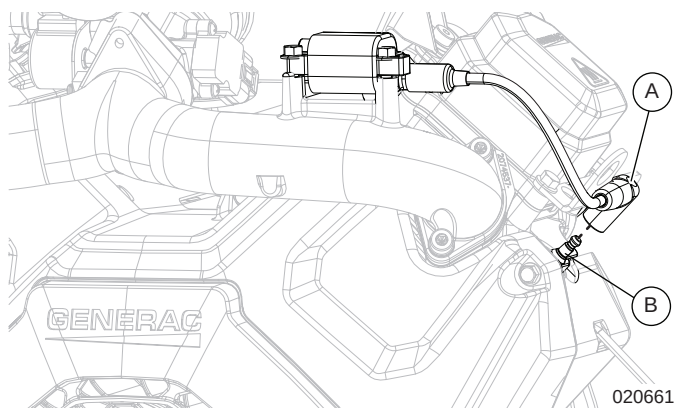


Figure 6-1. Remove Spark Plug Wires

2. See [Figure 6-2](#). Remove four fasteners (C) and the ignition coils (D, E) from mounting position on intake manifold.

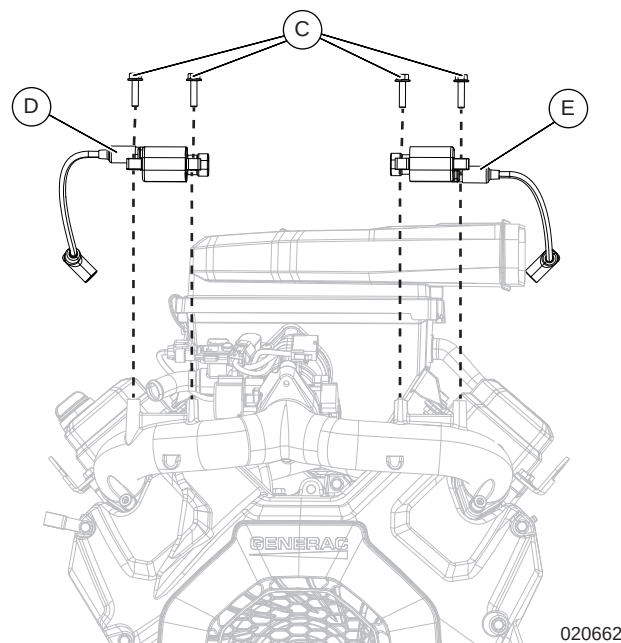


Figure 6-2. Remove Ignition Coils

3. Identify the ignition coils for cylinder one (E) and for cylinder two (D) to install in this same location during assembly.

Removing Spark Plugs and Starter

Proceed as follows to remove spark plugs and starter:

1. See [Figure 6-3](#). Remove spark plug (A) from each cylinder head.

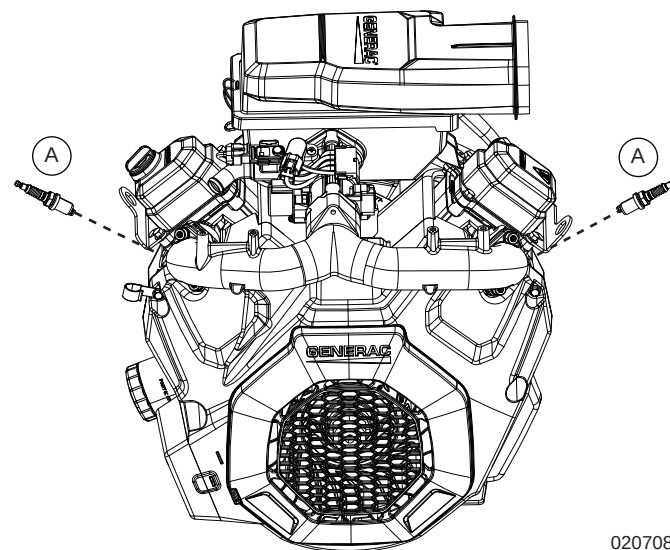


Figure 6-3. Remove Spark Plugs

2. See [Figure 6-4](#). Remove two starter fasteners (B), two lock washers (C), and starter motor (D) from crankcase.

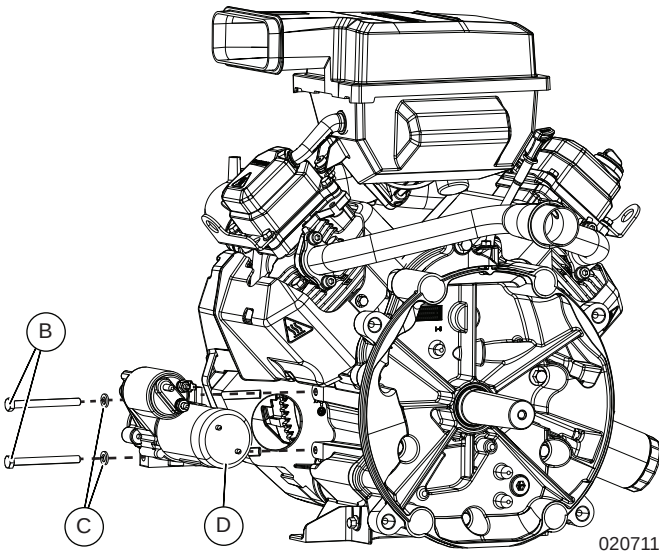


Figure 6-4. Remove Starter

Removing Air Box

Proceed as follows to remove air box:

1. See [Figure 6-5](#). Lift up and release hasp (A) securing air box cover (B).

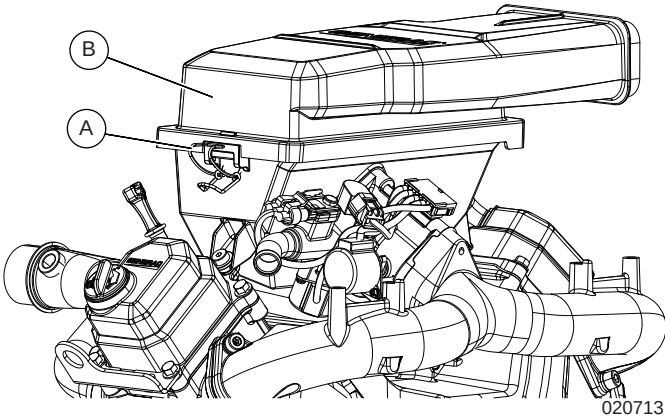


Figure 6-5. Release Hasp Securing Cover

2. See [Figure 6-6](#). Tip air box cover (B) away from hatch to release air box cover from connecting tabs on opposite side.

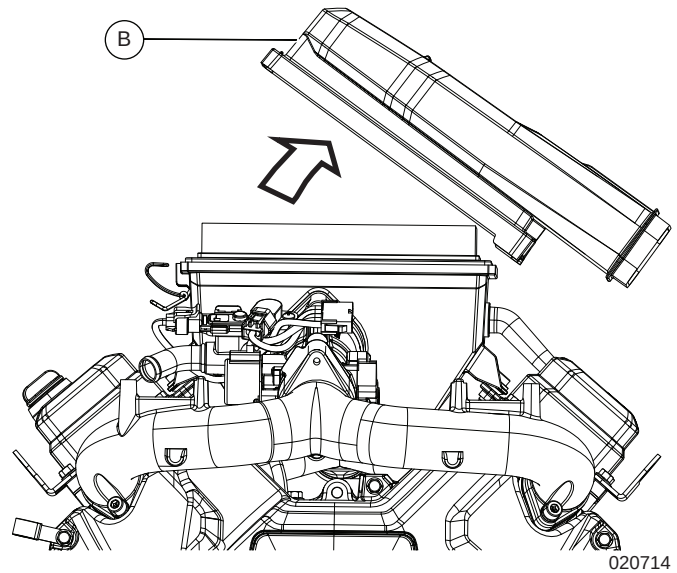


Figure 6-6. Remove Airbox Cover

3. See [Figure 6-7](#). Lift filter media (C) vertically to remove from air box base (D).

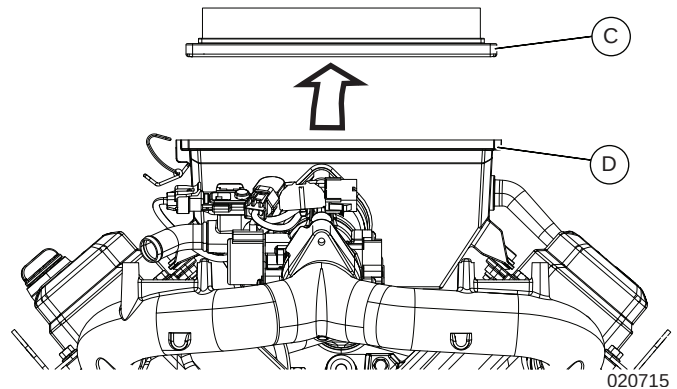


Figure 6-7. Remove Filter Media

4. See [Figure 6-8](#). Remove the two bolts (E) from inside the air box base (D) attaching air box to mixer.

NOTE: Air box mixer is now free from bolted mounting and caution should be taken to not damage air box mixer when removing the remainder of the air box assembly.

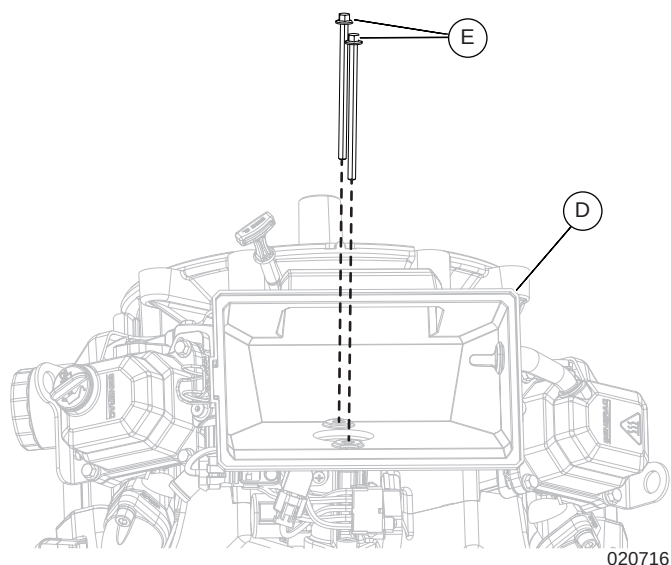


Figure 6-8. Remove Air Box Mixer Bolts

5. See [Figure 6-9](#). Remove air box base mounting bolts (F) from cylinder head and gently dislodge breather hose (G) from air box. Take care not to damage breather hose when removing from air box.

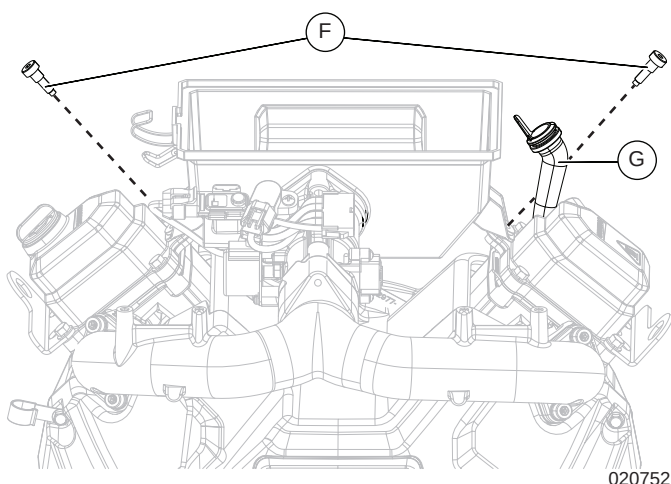


Figure 6-9. Remove Airbox Bolts and Breather Hose

6. See [Figure 6-10](#). Lift air box base (D) assembly from engine.

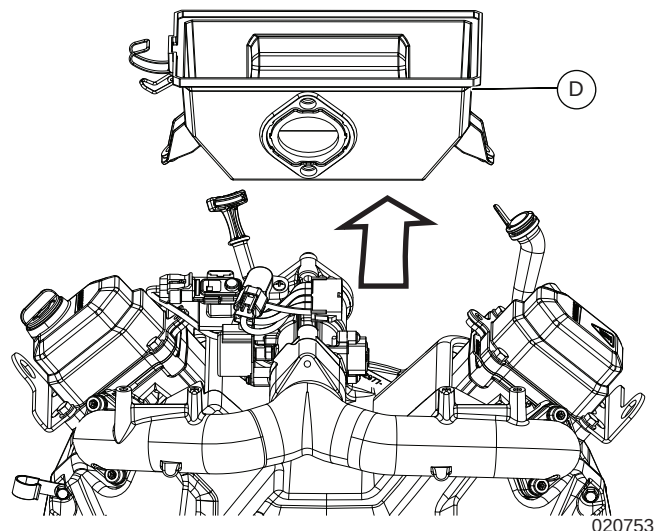


Figure 6-10. Remove Air Box Base From Engine

Removing Breather Hose From Rocker Cover

Proceed as follows to remove breather hose:

See [Figure 6-11](#). Gently remove breather hose (A) from rocker cover (B) facing the blower side of engine. Take extra care in removal to not damage hose.

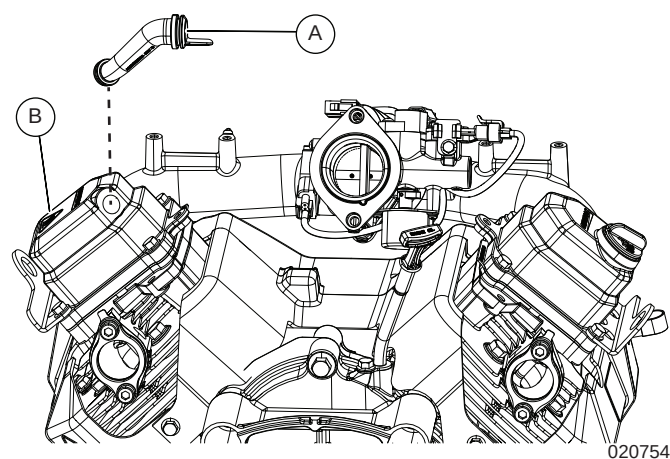


Figure 6-11. Remove Breather Hose

Removing Mixer Assembly

Proceed as follows to remove mixer assembly:

1. See [Figure 6-12](#). Remove O-ring (A) from groove on face of mixer (B) where it mates the air box bottom.

NOTE: There will no longer be bolts attaching mixer to engine once air box is removed from engine. Take care not to damage mixer or O-ring upon disassembly.

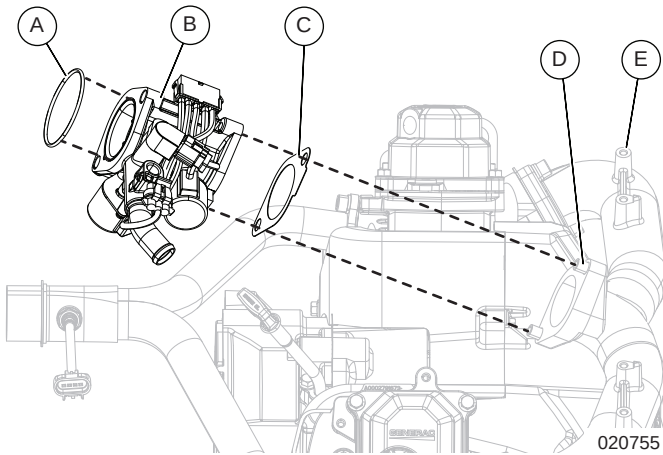


Figure 6-12. Remove Mixer Assembly

2. Carefully remove mixer (B) from alignment inserts (D) on engine intake manifold (E).
3. Remove gasket (C) from intake manifold. Take care not to damage gasket.

Disassembling Intake and Exhaust Components

Proceed as follows to disassemble intake and exhaust components:

1. See [Figure 6-13](#). Remove four exhaust manifold bolts (A) and washers (B).

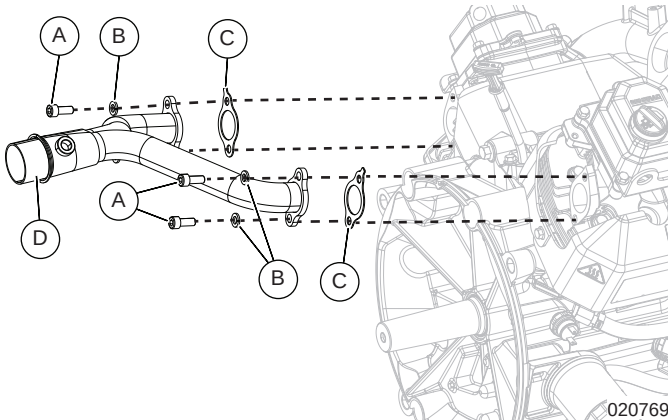


Figure 6-13. Remove Exhaust Manifold

2. Remove exhaust manifold (D).
3. Discard old gaskets (C). Always use new gaskets when assembling.
4. See [Figure 6-14](#). Remove four bolts (E) and intake manifold (F) from the cylinder heads. Discard intake manifold gaskets (G).

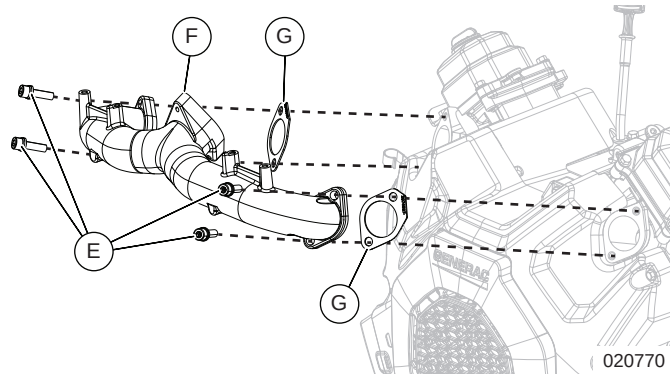


Figure 6-14. Remove Intake Manifold

Removing External Oil Components

Removing External Lubrication System Components

Proceed as follows to remove the external lubrication system components:

1. See [Figure 6-15](#). Cut off the crimped ear of the stepless ear clamp (A) to remove it from oil drain hose assembly.

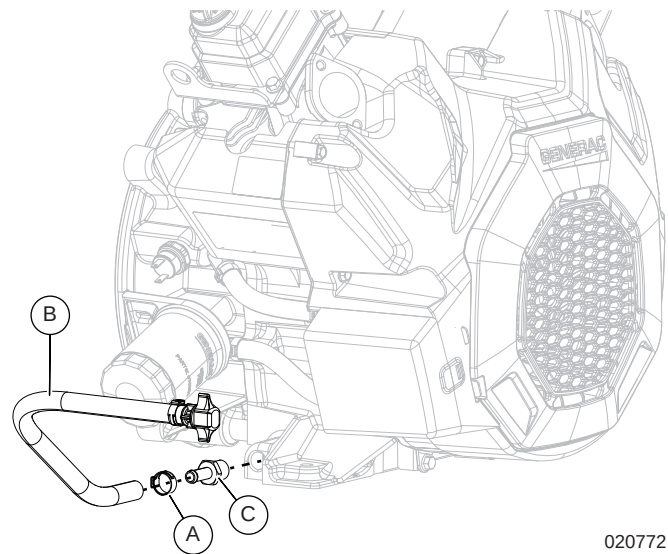


Figure 6-15. Remove Oil Drain Hose

NOTE: Factory installed stepless ear clamps are single use and must be replaced after removal.

2. Remove oil drain hose assembly (B) from barbed fitting.
3. Remove barbed fitting (C) from crankcase.
4. See [Figure 6-16](#). Remove oil filter (F) and low oil pressure switch (G) from oil filter mount.

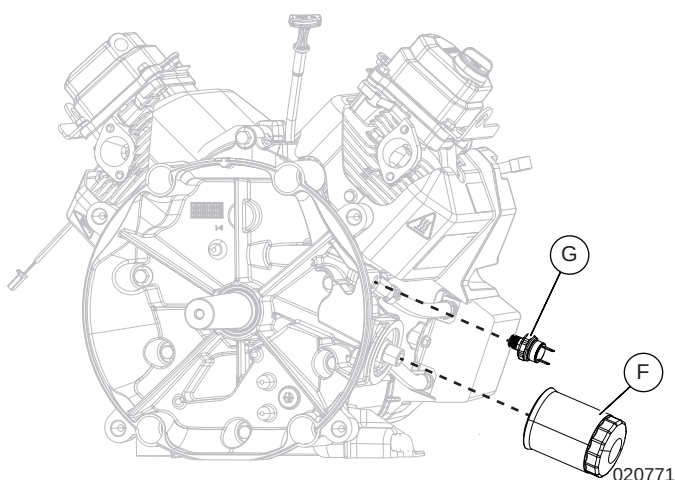


Figure 6-16. Remove Oil Filter and Low Oil Pressure Switch

Removing Oil Level Switch

Proceed as follows to remove the oil level switch.

1. See [Figure 6-17](#). Remove screw (A) fastening the oil level switch (B).

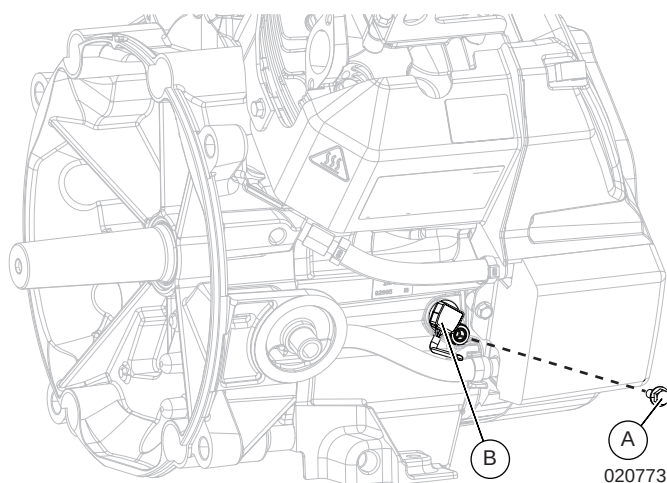


Figure 6-17. Remove Screw Fastening Switch

2. See [Figure 6-18](#). Carefully rotate and guide oil level switch (B) from port.

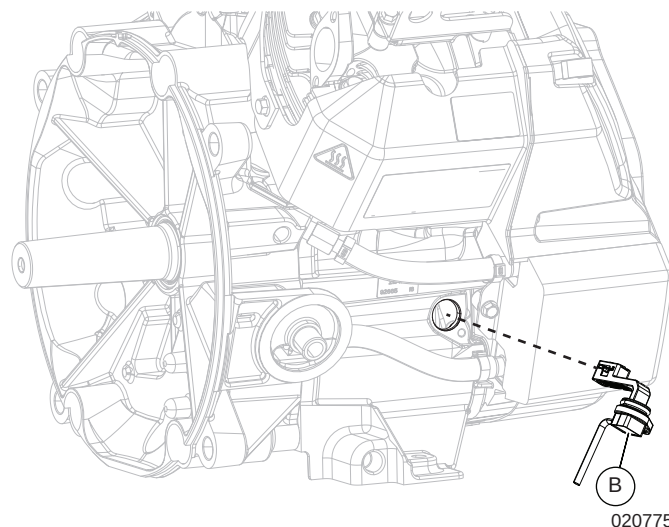


Figure 6-18. Remove Switch From Port

Removing Oil Cooler

Proceed as follows to remove oil cooler:

1. See [Figure 6-19](#). Remove screw (A) fastening oil cooler cover (B) and oil cooler (C) to blower housing.

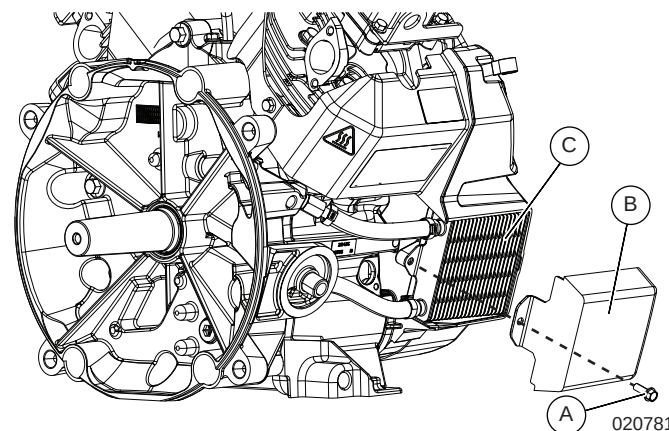


Figure 6-19. Remove Oil Cooler Screw and Cover

2. Remove oil cooler cover.
3. See [Figure 6-20](#). Cut off ears (D) on stepless ear clamps (E) to allow removal of hoses from barbed fittings. Remove stepless ear clamps.

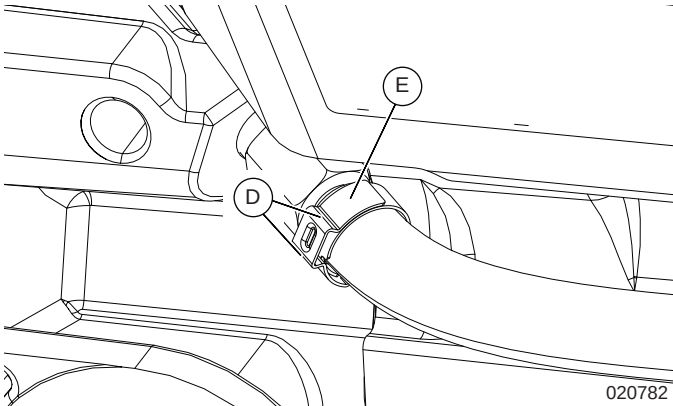


Figure 6-20. Cut Off Oetiker Clamp Ears

NOTE: Factory installed stepless ear clamps are single use and must be replaced after removal.

4. See [Figure 6-21](#). Remove rubber hoses (F) from barbed fittings (G).

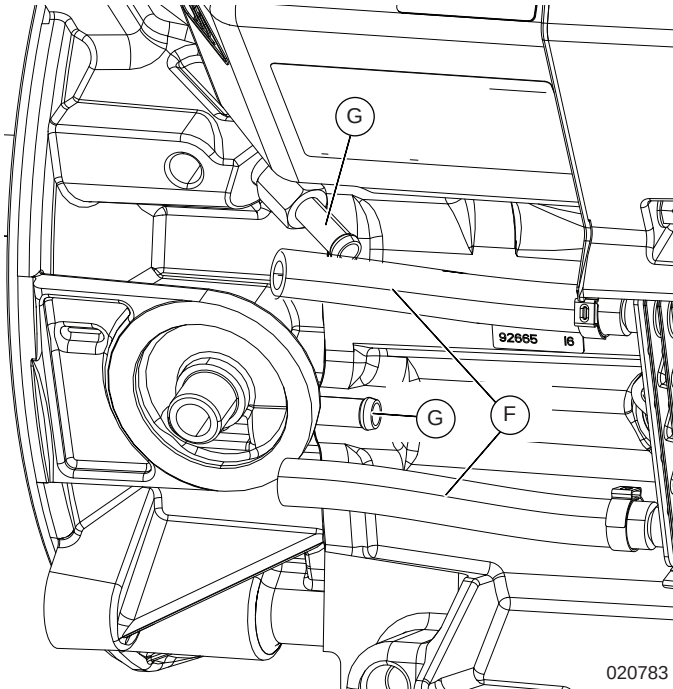


Figure 6-21. Remove Rubber Hoses

5. See [Figure 6-22](#). Move oil cooler away from blower housing to free cooler from mounting slots (H) in blower housing. Remove cooler (B) from engine.

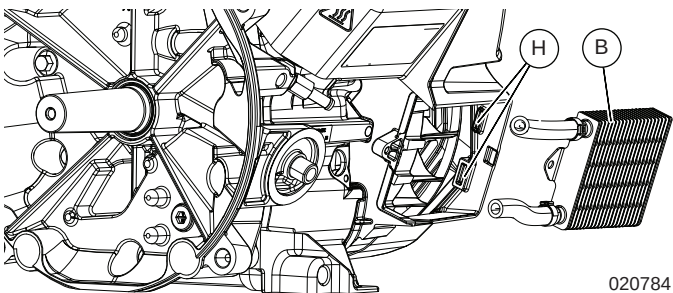


Figure 6-22. Remove Oil Cooler

6. See [Figure 6-23](#). Remove oil cooler barbed fittings (J) from engine.

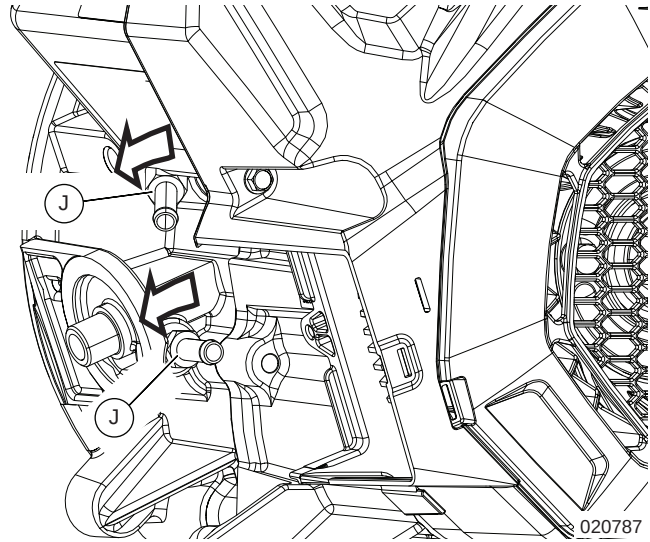


Figure 6-23. Remove Oil Cooler Barbed Fittings

Removing Oil Dipstick

Proceed as follows to remove oil dipstick:

1. See [Figure 6-24](#). Remove bolt (A) retaining oil dipstick tube.

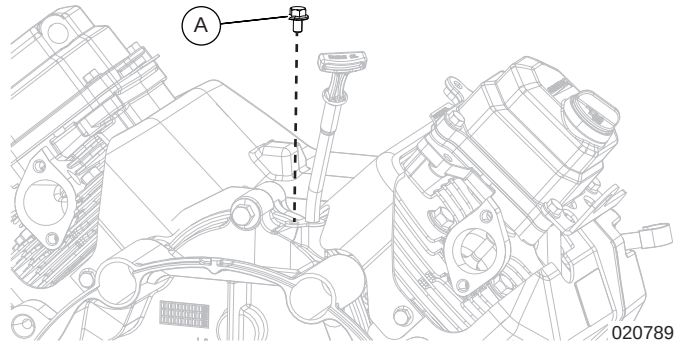


Figure 6-24. Remove Dipstick Retaining Bolt

2. See [Figure 6-25](#). Remove oil dipstick (B) from engine.

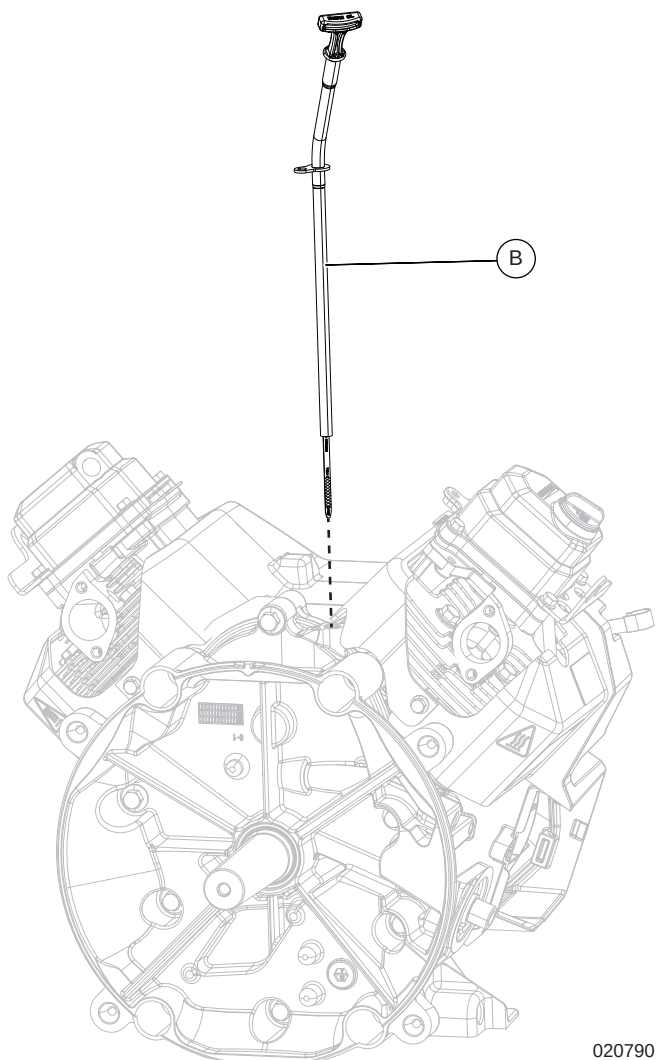


Figure 6-25. Remove Oil Dipstick

Removing Cylinder Wrappers and Blower Housing

Proceed as follows to remove cylinder wrappers and blower housing:

1. See [Figure 6-26](#). Remove five fasteners (A), oil drain J-clip (B), and blower housing (C).

NOTE: Disengage the plastic tab, located on the bottom of the housing, before removal.

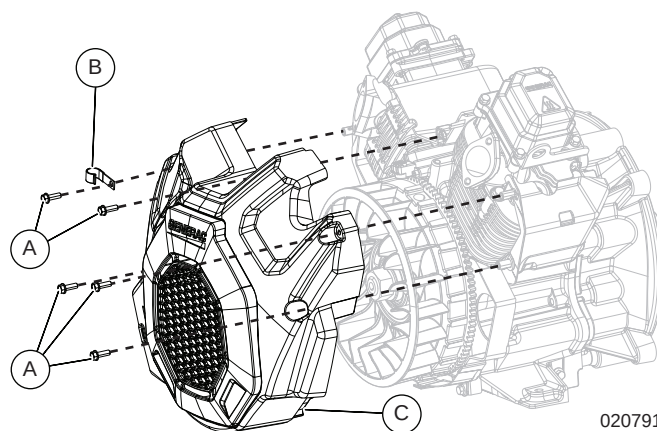


Figure 6-26. Remove Blower Housing

2. See [Figure 6-27](#). Remove two fasteners (D) and side wrappers (E, F). Remove top wrapper (G).

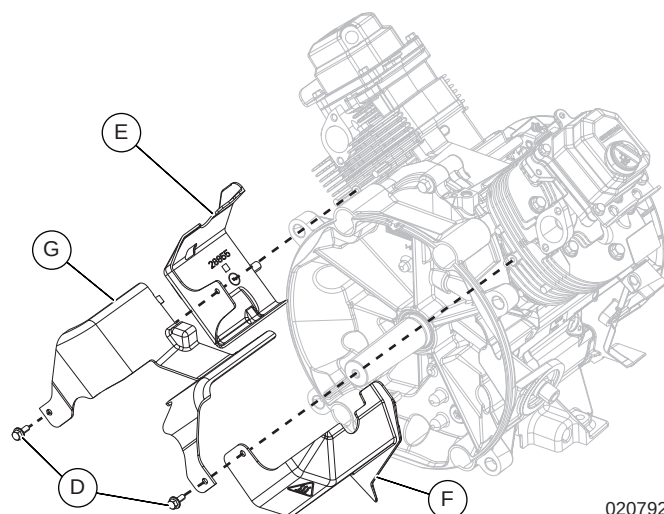


Figure 6-27. Remove Cylinder Wrappers

Removing Flywheel

Proceed as follows to remove flywheel:

1. See [Figure 6-28](#). Remove two fasteners (A), fan plate (B), and fan (C) from flywheel (D).

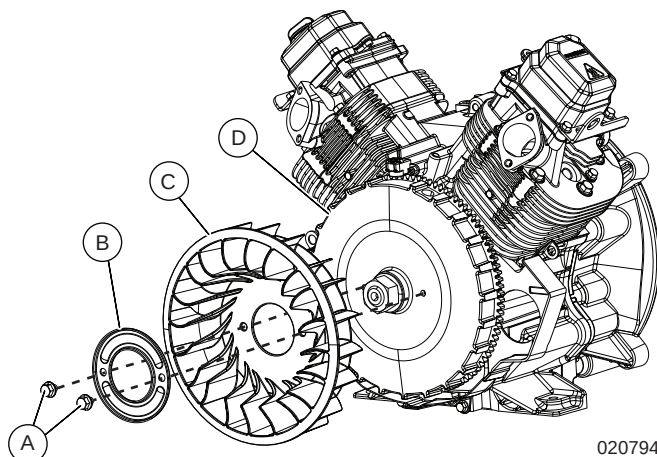


Figure 6-28. Remove Fan

2. See [Figure 6-29](#). Remove bolts (E) securing crankshaft position sensor (F). Remove sensor from engine.

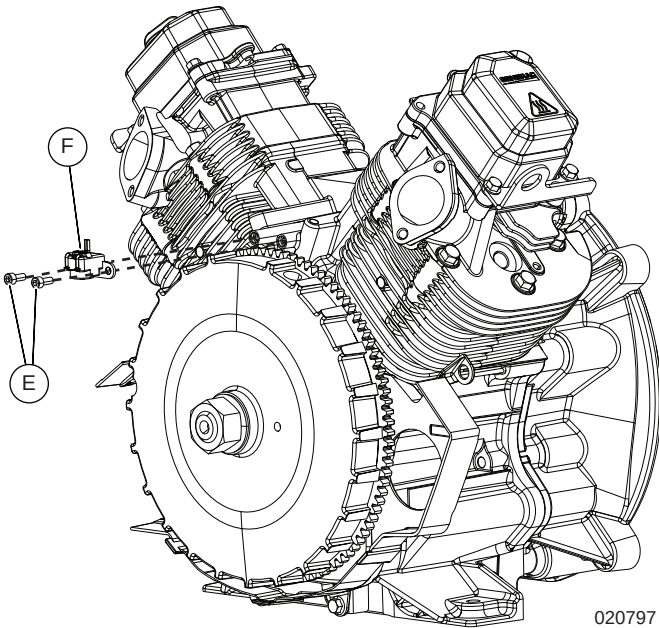


Figure 6-29. Remove Crankshaft Position Sensor

3. See [Figure 6-30](#). Loosen flywheel nut (G) until it is flush with end of crankshaft threads.

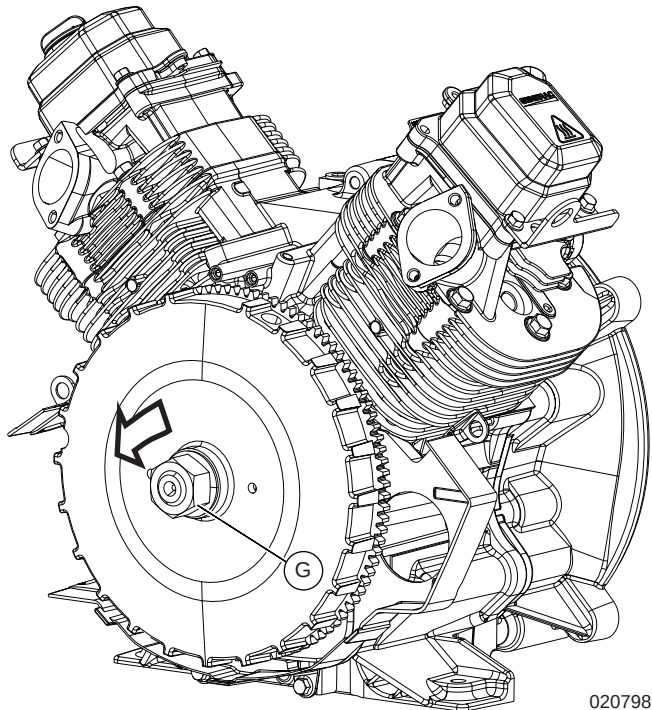


Figure 6-30. Loosen Flywheel Nut

4. See [Figure 6-31](#). Install flywheel puller (H) using two M8 x 1.25 mm bolts (J).

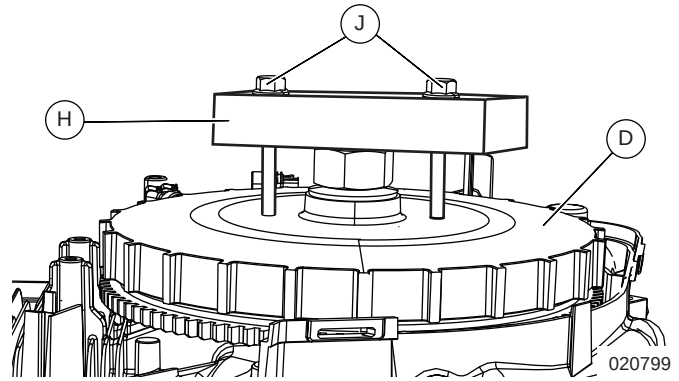


Figure 6-31. Install Flywheel Puller

5. Tighten puller bolts (G) until flywheel (D) loosens.
6. Remove flywheel puller.
 - Do NOT over tighten flywheel puller bolts.
 - DO NOT strike flywheel with a hard object or a metal tool. Doing so may cause flywheel to shatter in operation. Always use approved flywheel removal tools.
7. See [Figure 6-32](#). Remove nut (K), washer (L), flywheel (D), and woodruff key (M) from crankshaft.

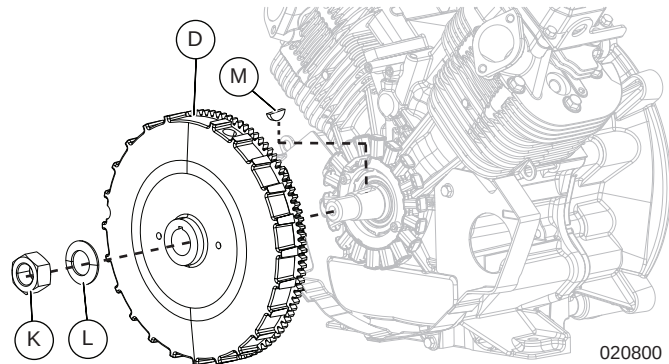


Figure 6-32. Remove Flywheel

8. See [Figure 6-33](#). Remove four fasteners (K) and backing plate (L) from engine.

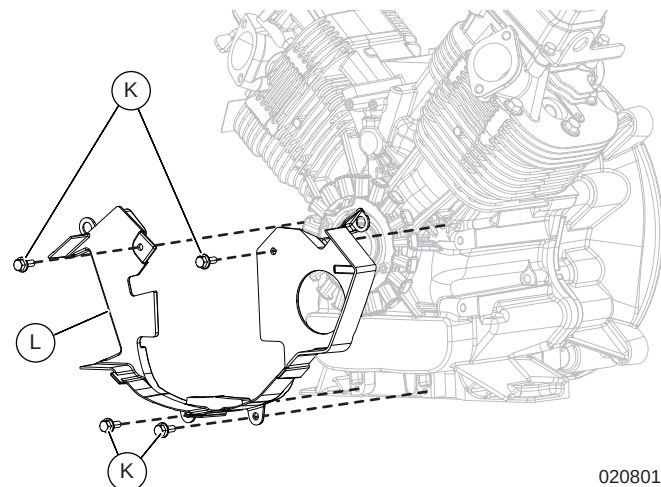


Figure 6-33. Remove Backing Plate

Removing Battery Charger Stator

Proceed as follows to remove battery charger stator:

1. Disconnect stator wiring from harness.
2. See [Figure 6-34](#). Remove the two bolts (A) securing the battery charger stator (B).

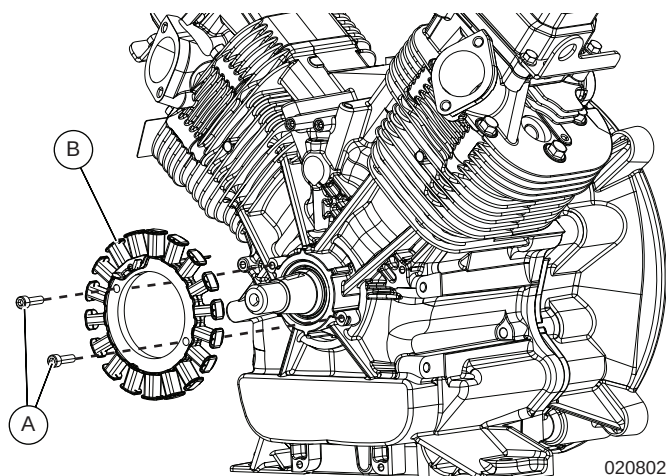


Figure 6-34. Remove the Bolts

3. Remove battery charger stator.

Removing Rocker Covers

Proceed as follows to remove rocker covers:

1. See [Figure 6-35](#). Remove the four fasteners (A), lifting bracket (B), and rocker cover (C) from cylinder head.

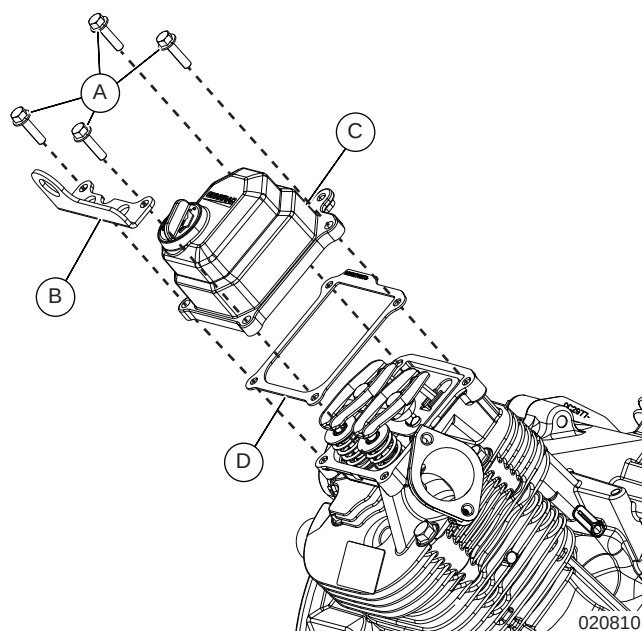


Figure 6-35. Remove Rocker Cover

2. Remove and discard gasket (D).
3. Repeat steps 1 and 2 for other rocker cover.

Removing Pushrods and Rocker Arms

Proceed as follows to remove pushrods and rocker arms:

1. See [Figure 6-36](#). Remove rocker arm pivot studs (A), rocker arms (B), and pushrod guide plate (C) from cylinder head.

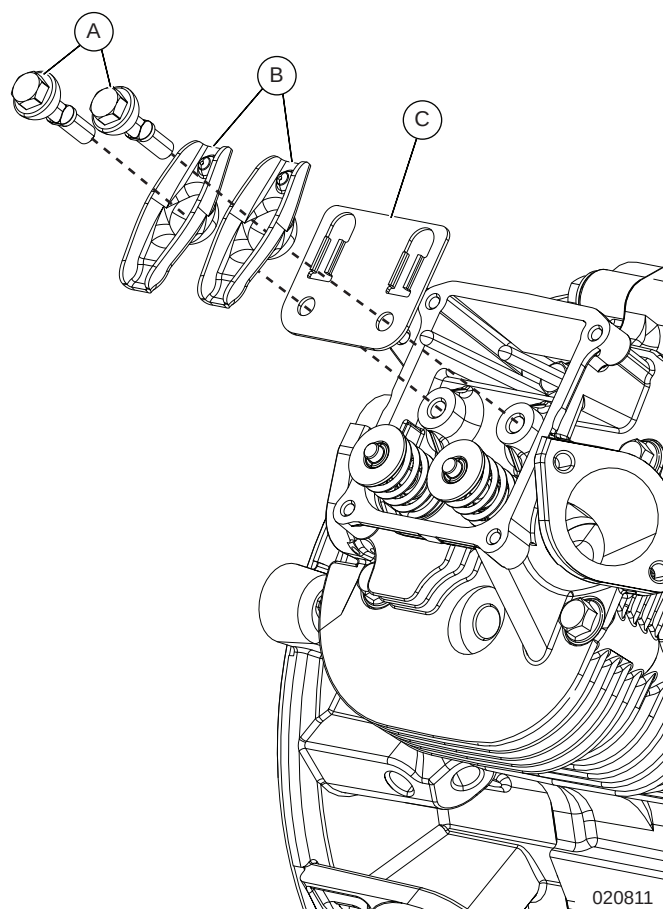


Figure 6-36. Remove Rocker Arms

2. See [Figure 6-37](#). Remove the two pushrods (D).

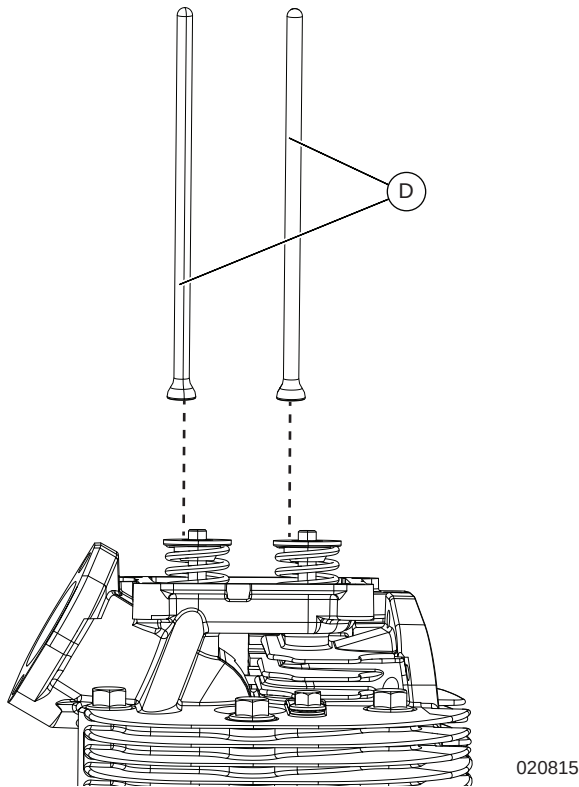


Figure 6-37. Remove Pushrods

3. Repeat cylinder head disassembly steps 1 and 2 for other cylinder head.

Removing Cylinder Heads

Proceed as follows to remove cylinder heads:

1. See [Figure 6-38](#). Remove the five cylinder head bolts (A), washers (B), cylinder head assembly (C), and gasket (D) from engine long block.

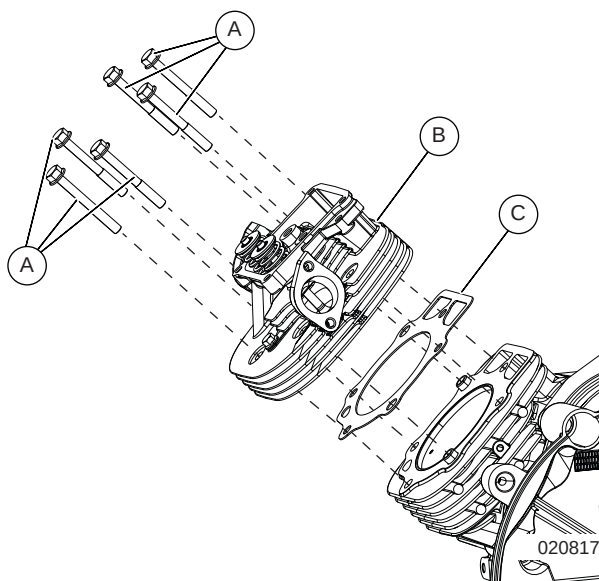


Figure 6-38. Remove Cylinder Head Assembly

2. Remove and discard head gasket (D).

3. See [Figure 6-39](#). Remove the cylinder head bolt dowels (E) from cylinder head assembly.

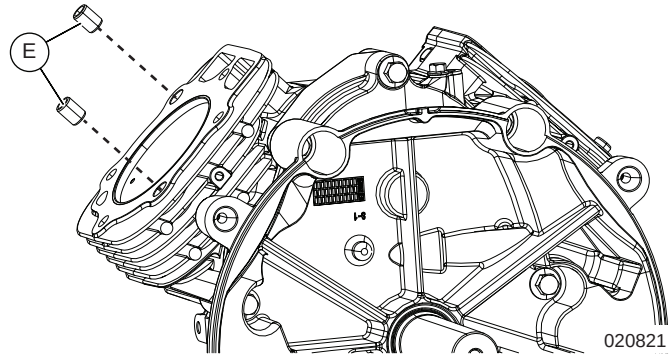


Figure 6-39. Remove Cylinder Head Bolt Dowels

4. Repeat cylinder head disassembly steps 1–3 for other cylinder head.

Disassembling Cylinder Head



Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury.

(W000556)



Personal Injury. Properly compress valve springs using a valve spring compressor. Failure to do so could cause the spring to eject, and could result in serious injury.

(W000193)

Proceed as follows to disassemble cylinder head:

1. See [Figure 6-40](#). Use a valve spring compressor to compress valve spring retainer (A) and valve spring (B) sufficiently to remove the two valve spring keepers (C) from valve stem (D).

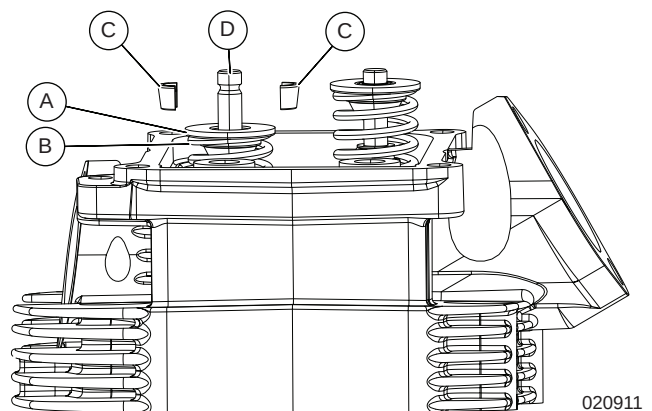


Figure 6-40. Remove Valve Spring Keepers

2. See [Figure 6-41](#). Remove valve spring retainer (A) and valve spring (B).

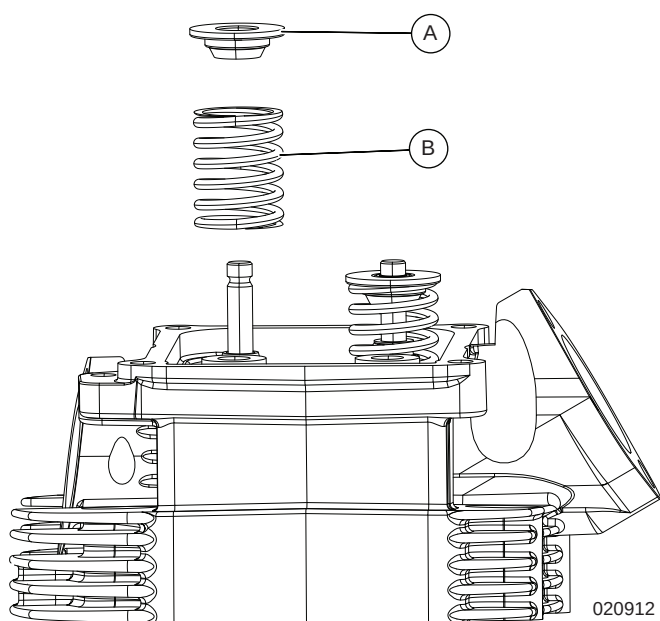


Figure 6-41. Remove Valve Spring

IMPORTANT NOTE: Spring compressor is under a large amount of tension. Take special care when releasing tension.

NOTE: Valve will be loose in cylinder head. Take care not damage valve by allowing valve to drop out of cylinder head.

3. See [Figure 6-42](#). Remove intake valve (E) from combustion chamber side.

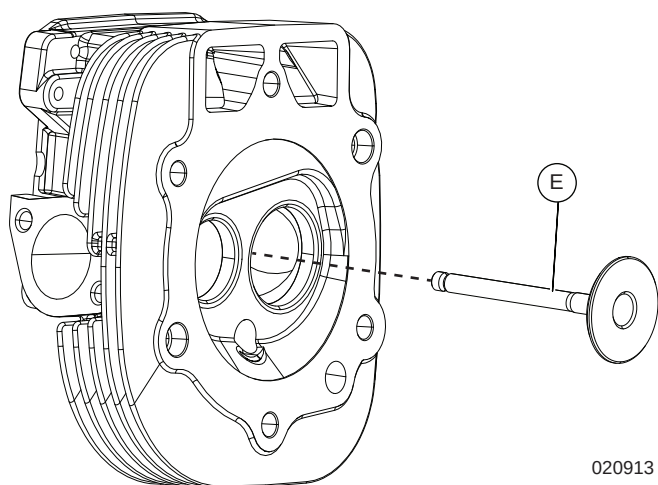


Figure 6-42. Remove Intake Valve

4. See [Figure 6-43](#). Remove valve stem seal (F) from cylinder head intake guide.

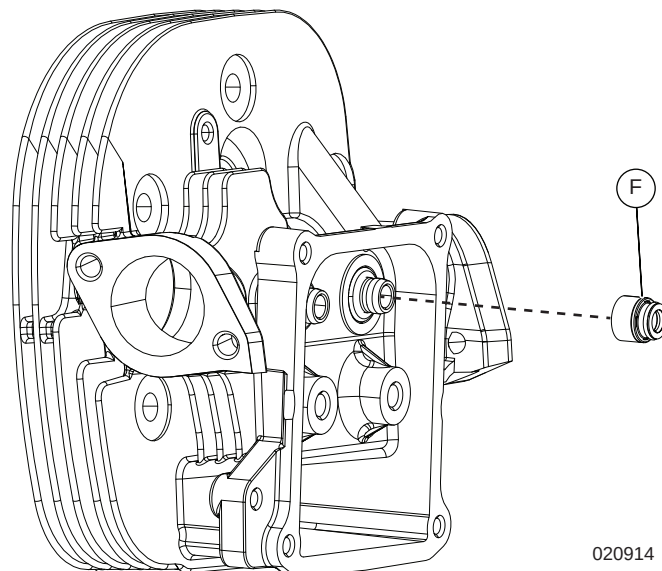


Figure 6-43. Remove Valve Stem Seal

5. See [Figure 6-44](#). Repeat steps 1–4 to remove exhaust valve (G) from cylinder head assembly.

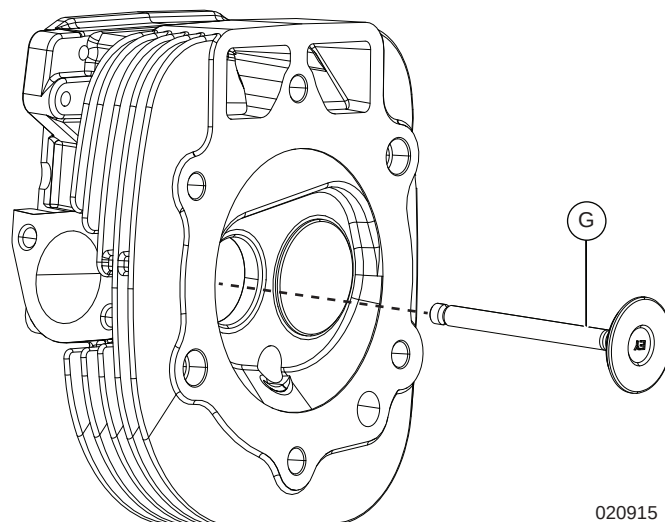


Figure 6-44. Remove Exhaust Valve

6. Repeat cylinder head disassembly steps 1–5 for second cylinder head.

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Section 7: Ignition

Ignition Check

The ignition system is controlled by the ECM. The coils are powered by the IR and fuse F4. The IR is controlled by the ECM.

In order to generate a fault code, the following conditions may apply:

- The ECM will check if ignition coil current is above the threshold when coil is firing.
- The ECM will check if ignition coil current is below the threshold when the coil is firing.
- The ECM will check if ignition coil current is detected when coil is not firing.

Inspecting Ignition

Inspect ignition system for correct function if engine runs but misfires during operation. Proceed as follows to inspect ignition system:

1. See [Figure 7-1](#). Install a spark tester (A) (Generac part number 0C5969) between spark plug lead (B) and each spark plug (C), as shown. Set spark tester to a 10 kV gap. Test each plug separately if only one spark tester is available.

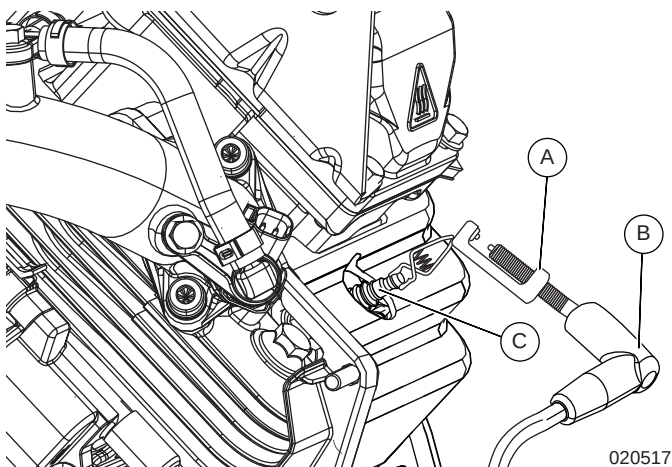


Figure 7-1. Spark Tester on Engine (Typical)

2. Start engine and verify if a steady spark is visible in the window of the tester. A lack of spark or misfire will be readily apparent when the engine is running, indicating further investigation of the ignition system is required.
3. If spark is steady for both cylinders but engine still misfires or will not run, inspect for a fouled spark plug. Verify fuel is present in the fuel system and power is being supplied to the fuel shutoff solenoid.

4. If there is no spark, connect spark tester to engine block (ground) and crank engine. If spark is present, spark plugs are damaged.

Removing Ignition Coils

Proceed as follows to remove the ignition coils for testing:

1. See [Figure 7-2](#). Remove spark plug wires (A) from spark plugs (B).

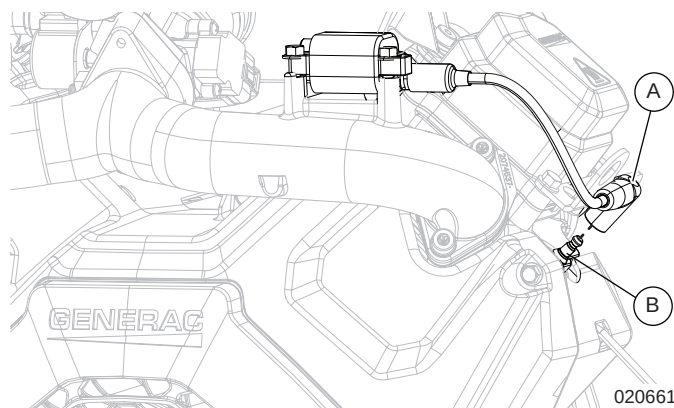


Figure 7-2. Remove Spark Plug Wires

2. See [Figure 7-3](#). Remove bolts (C) retaining ignition coils (D, E) to intake manifold and remove ignition coils.

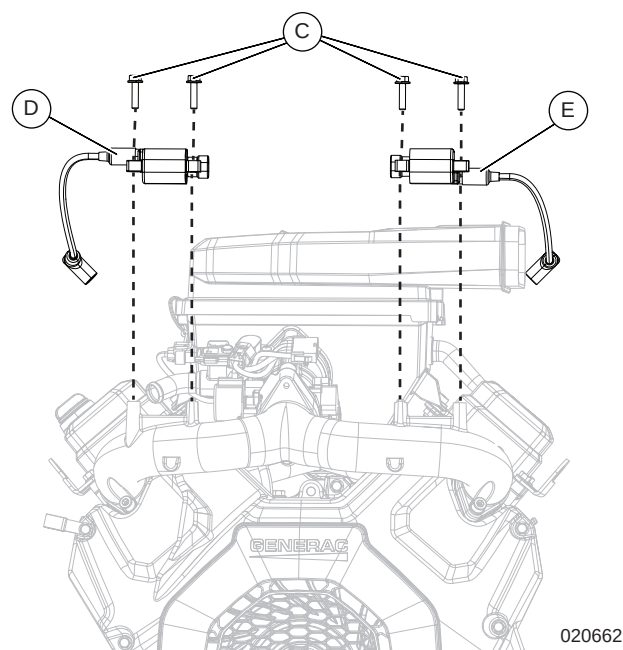


Figure 7-3. Remove Ignition Coils From Intake Manifold

3. Inspect coil wire lead to verify it has not worn through allowing wiring to ground to backing plate.

If contact is identified, repair/replace wire and check for spark.

Testing Ignition Coil

If ignition inspection indicates one or both of the ignition coils are not delivering a strong consistent spark, proceed as follows to test ignition coil:

1. Disconnect the electrical connection and the primary wire from the coil.

NOTE: It may be necessary to remove the coil from the valve cover.

2. Set digital multimeter (DMM) to measure resistance.
3. See [Figure 7-4](#). Attach positive test lead to primary wire terminal 451 to measure resistance across coil number 1 and wire 452 to measure resistance across coil number 2.

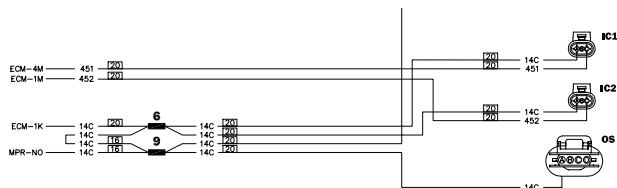


Figure 7-4. Ignition Coil Wiring Diagram

4. Attach negative test lead to wire terminal 14C to measure resistance on either coil.
5. See [Figure 7-5](#). Measure resistance across coil. Resistance on primary side should read 1 ohms $\pm$ 15%.

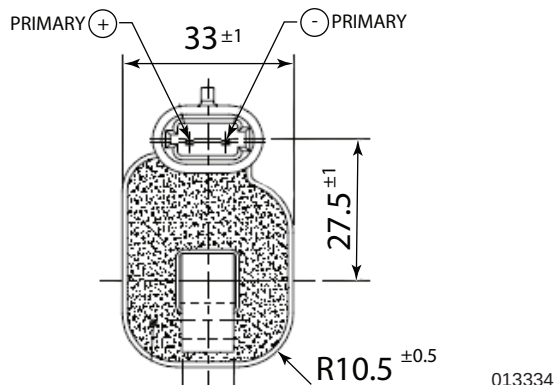


Figure 7-5. Ignition Coil

6. If coil is out of spec, replace coil.

Checking Wire Harness

If ignition does not have spark, disconnect bullet connection from ignition grounding stud on engine and check again. If ignition now has spark, check wire harness for pinched wires, broken insulation, or (if equipped) faulty diode. If there is no spark, proceed to [Removing Ignition Coil](#).

Removing and Installing Wire Harness

Proceed as follows to remove and install wire harness:

1. Remove spark plug leads.
2. Remove intake manifold and cover intake ports with a shop towel.
3. Remove rotating screen and blower housing.
4. Disconnect stop switch wire(s) at ignition coil(s).
5. Reverse order of steps above to install new wire harness.

Flywheel

The flywheel is a weighted disc which helps keep the crankshaft rotating between power strokes and provides a reference for the crank position sensor to set the ignition timing.

Removing Flywheel and Fan

See [Removing Flywheel](#) to correctly remove fan and flywheel from engine.

Installing Flywheel and Fan

Proceed as follows to install flywheel and fan:

1. See [Figure 7-6](#). Install woodruff key (A) into slot in crankshaft (B).

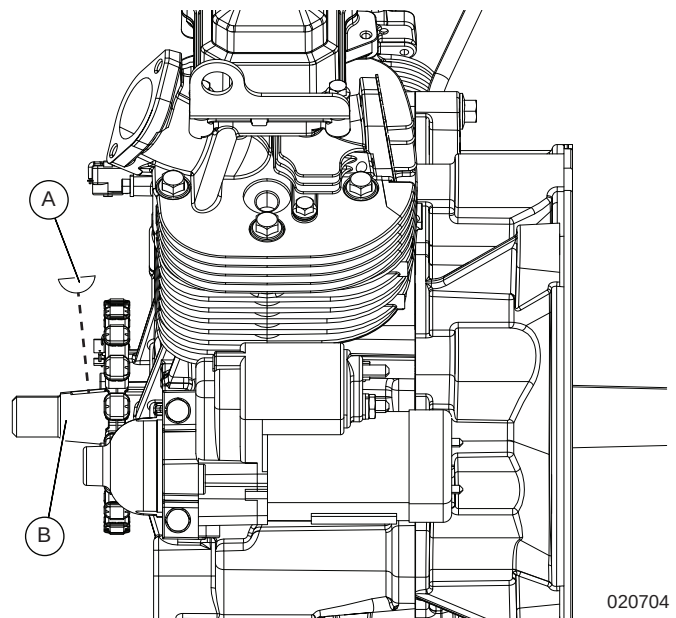


Figure 7-6. Install Woodruff Key

2. Verify both crankshaft and flywheel tapers are clean and free of oil and debris.
3. See [Figure 7-7](#). Align keyway and install flywheel (C) on crankshaft.

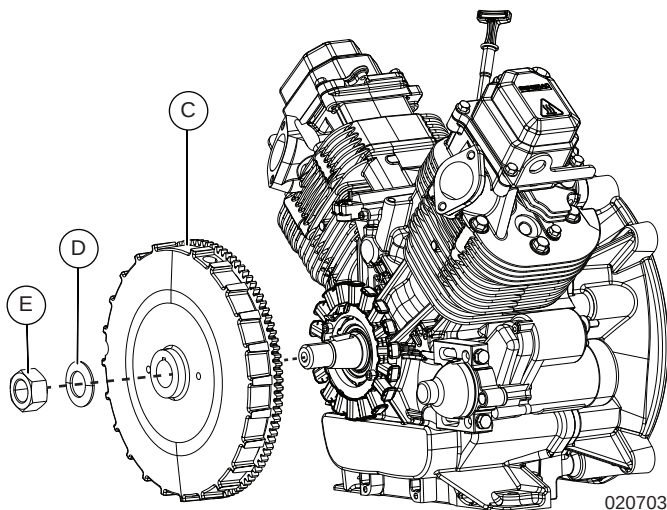


Figure 7-7. Install Flywheel, Washer, and Nut

4. Attach flywheel to crankshaft using flywheel washer (D) and flywheel nut (E).
5. See [Figure 7-8](#). Tighten nut to 105–129 ft-lb (142–175 Nm).

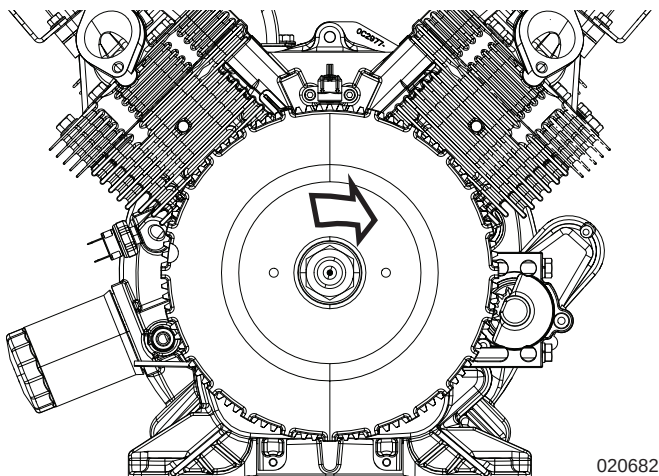


Figure 7-8. Tighten Flywheel Nut

6. See [Figure 7-9](#). Install fan (F) and fan plate (G) to flywheel using two fasteners (H).

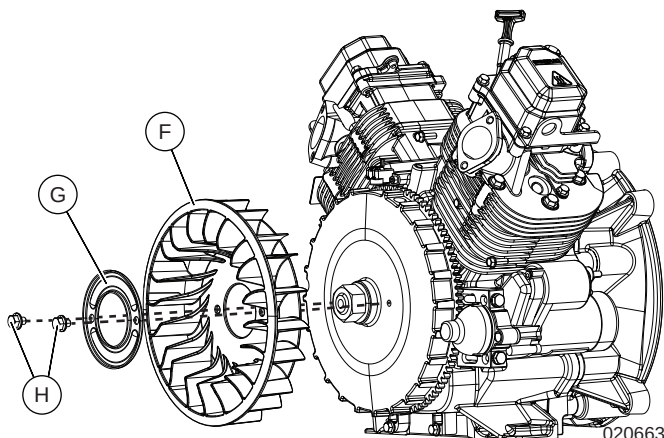


Figure 7-9. Install Flywheel Fan

7. Tighten bolts to 16–20 ft-lb (22–27 Nm).

Installing Crankshaft Position Sensor

Proceed as follows to install crankshaft position sensor:

1. See [Figure 7-10](#). Install crankshaft position sensor (N) using fasteners (M).

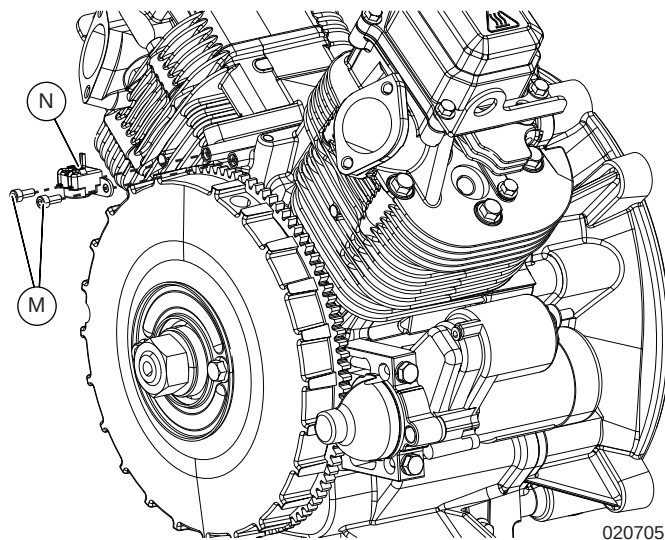


Figure 7-10. Install Crankshaft Position Sensor

2. Tighten fasteners to 53.1 - 57.5 in-lbs (6-6.5 Nm).

Checking Spark Plug Gap

Spark plug (s) should be cleaned and re-gapped periodically. Spark plug(s) should be replaced every 200 hours of operation or once annually, whichever comes first.

See [Figure 7-11](#). Set spark plug gap to 0.020 in (0.50 mm) using a wire-type feeler gauge.

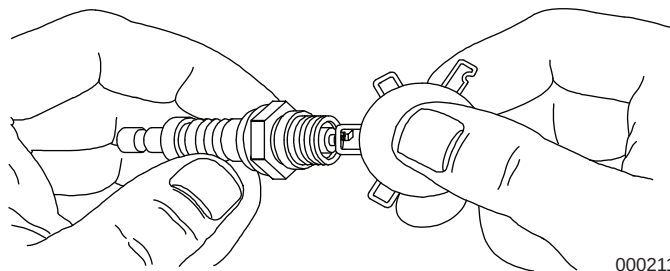


Figure 7-11. Check Spark Plug Gap

NOTE: For correct spark plug replacement, see owner's manual for the specific product. Always use original Generac replacement parts.

Installing Ignition Coils

Ignition coils mount to the intake manifold and only can be installed after installation of the intake manifold.

Proceed as follows to install ignition coils:

1. See [Figure 7-12](#). Install ignition coils (D, E) for cylinder one and cylinder two with spark plug lead pointing upwards.

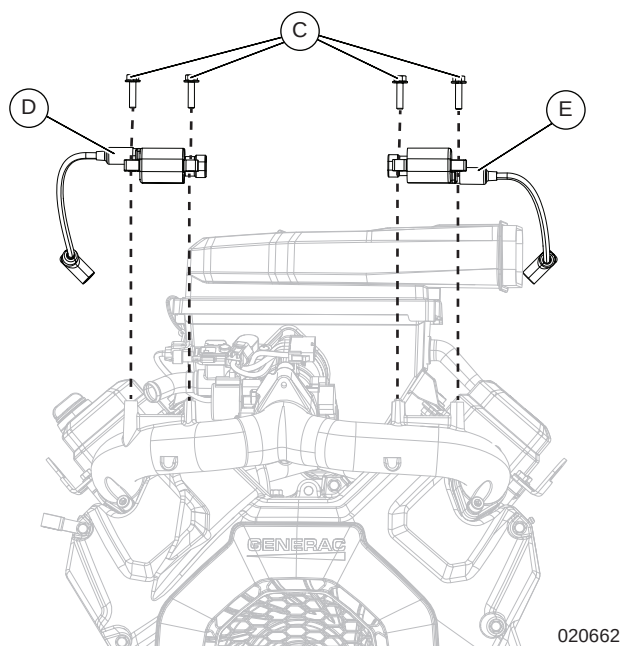


Figure 7-12. Install Ignition Coils

2. Fasten coils to crankcase using four fasteners (C) and tighten to 88.5 - 106.2 in-lbs (10-12 Nm)..
3. See [Figure 7-13](#). Install spark wires (A) to spark plugs (B).

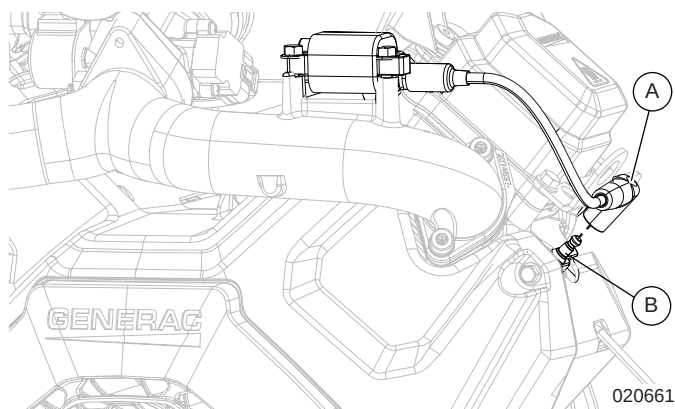


Figure 7-13. Install spark plug wires

Section 8: Sub-Assemblies

Gear Cover Sub-Assembly

Proceed as follows to install oil seal and check torque on gear cover assembly:

1. See [Figure 8-1](#). Press the crankshaft PTO oil seal (A) into the gear cover assembly (B) until seal is fully seated against bottom of bore.

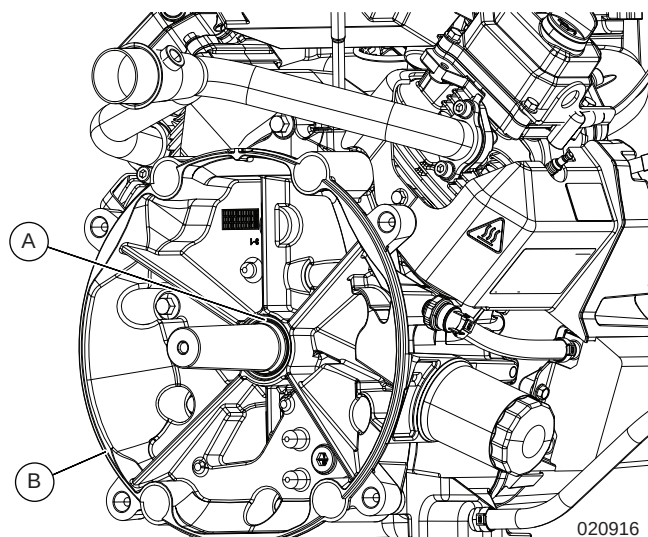


Figure 8-1. Install Oil Seal Into Gear Cover

2. See [Figure 8-2](#). Check gear cover torque using the correct sequence shown. Final torque on gear cover is 32.4–39.6 ft-lb (43.9–53.7) Nm.

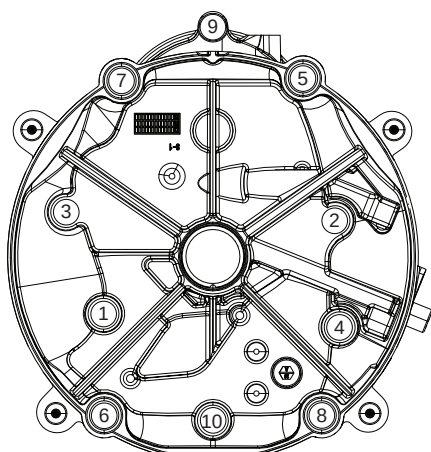


Figure 8-2. Gear Cover Torque Sequence

Cylinder Head Sub-Assembly

Proceed as follows to complete the cylinder head sub-assembly:

1. See [Figure 8-3](#). Press valve stem seal (F) onto cylinder head intake and exhaust guide.

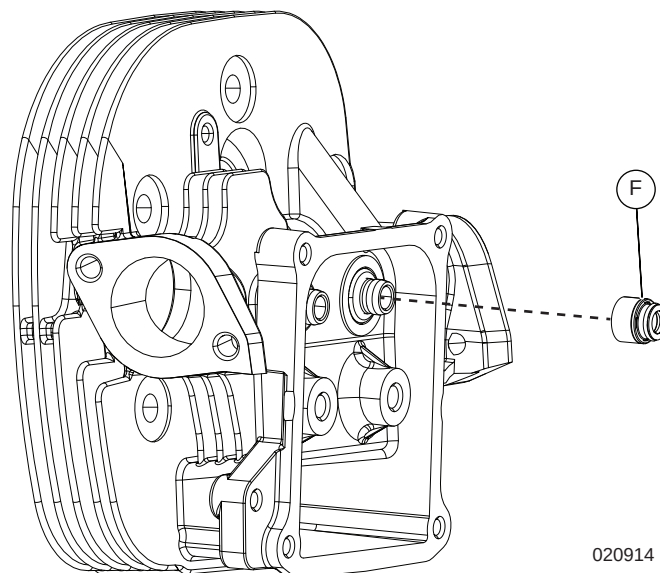


Figure 8-3. Install Valve Stem Seal

2. See [Figure 8-4](#). Apply assembly lube to the tip of intake valve (E) and insert valve into valve guide from combustion chamber side. Take care to not damage seal.

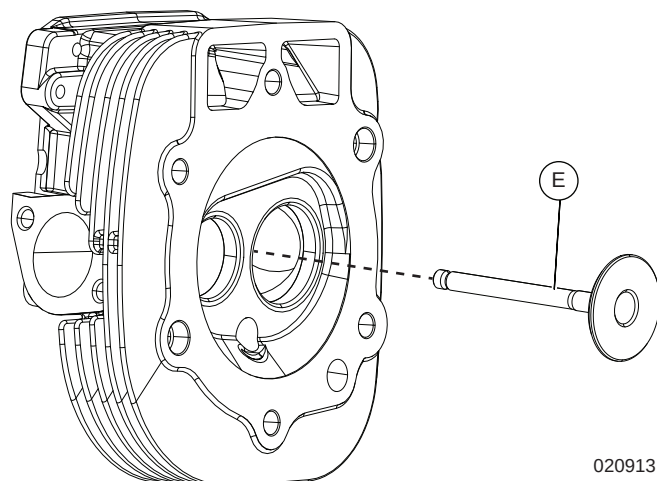


Figure 8-4. Install Intake Valve into Cylinder Head

3. See [Figure 8-5](#). Install valve spring (B) and spring retainer (A) onto cylinder head assembly.

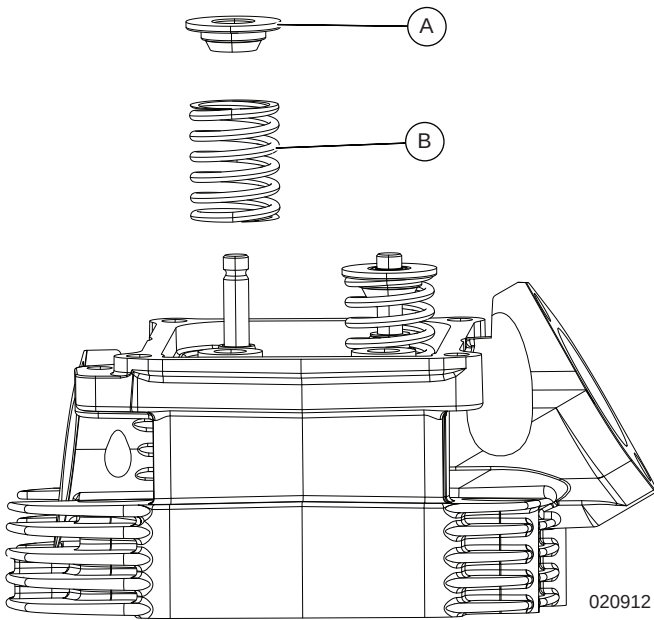


Figure 8-5. Install Valve Spring Assembly onto Valve

4. See [Figure 8-6](#). Use a valve spring compressor to compress the valve spring (B) sufficiently enough to install the two valve spring keepers (C). Verify keepers are locked into the slot of the valve stem.

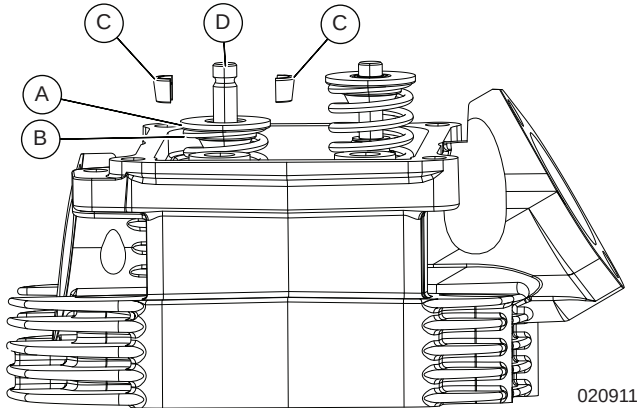


Figure 8-6. Install Valve Spring Keepers onto Valve

5. Release valve spring compressor. Verify keepers are firmly locked into slot on valve stem.
6. See [Figure 8-7](#). Repeat steps 1– 5 to install exhaust valve into cylinder head assembly.

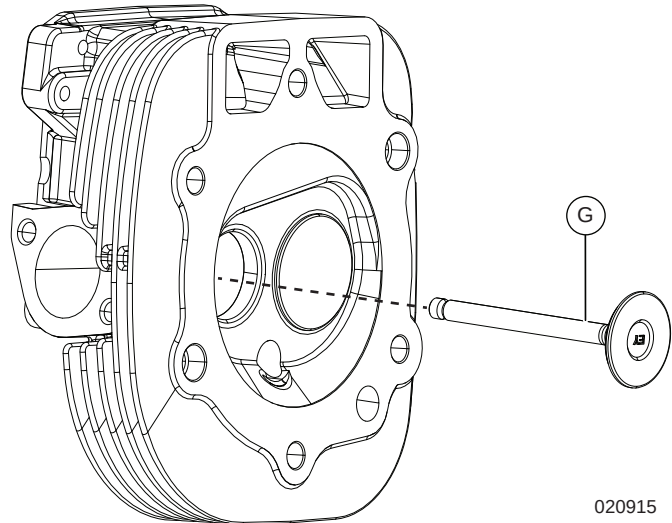


Figure 8-7. Install Exhaust Valve into Cylinder Head

7. Repeat cylinder head assembly steps 1–6 for the second cylinder head using the same components stated above.

Section 9: Cylinder Head and Valves

Inspecting and Repairing Cylinder Head

Proceed as follows to inspect and repair cylinder head:

1. See [Figure 9-1](#). Inspect condition of cylinder head. Verify all gasket material is removed from surfaces before inspecting.
 - a. Inspect cylinder head for cracks or damage.
 - b. Use a surface plate or straightedge and inspect cylinder head mounting surface for distortion.

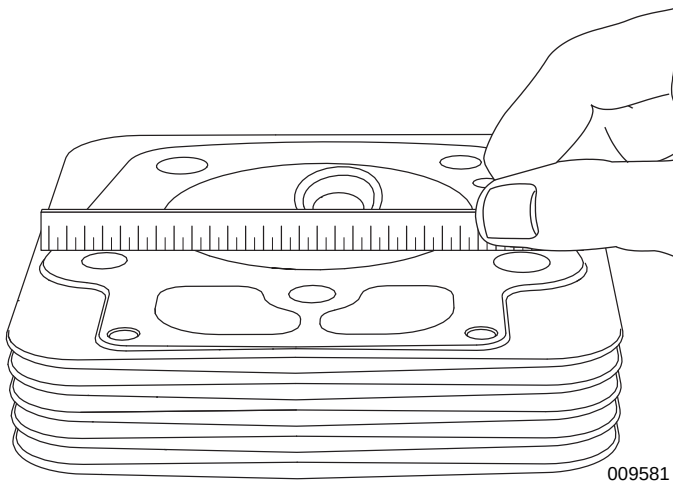


Figure 9-1. Check Cylinder Head Distortion

Cylinder head must be replaced if mounting surfaces are distorted more than 0.004 in (0.1 mm).

NOTE: Resurfacing cylinder head mounting surfaces is not recommended.

2. See [Figure 9-2](#). Clean valve guides and measure inner diameter using a split ball bore gauge. Replace head if either valve guide measures 0.278 in (7.06 mm) or more.

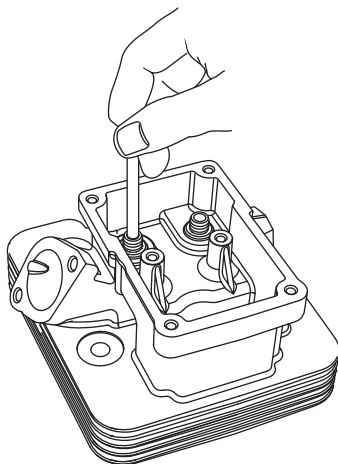


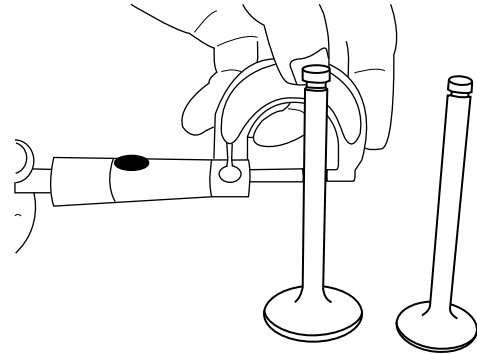
Figure 9-2. Measure Valve Guide Inner Diameter

NOTE: Valve guides are not replaceable.

Measuring Valves and Springs

Proceed as follows to measure valve stem diameter and valve springs:

1. See [Figure 9-3](#). Measure valve stem diameter at 1.600 in (40 mm) from end of valve.

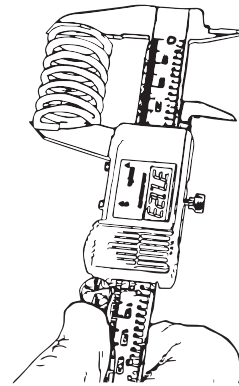


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Figure 9-3. Measure Valve Stem Diameter

Replace valves if stem diameter is less than 0.272 in (6.9 mm), or if total clearance between valve stem and valve guide exceeds 0.0047 in (0.12 mm).

2. See [Figure 9-4](#). Check valve springs for free length. Replace valve springs if free length is less than 1.437 in (36.5 mm).



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Figure 9-4. Check Valve Springs

Gasket Removal

Verify sealing surface is completely clean and free of old gasket material before installing a new gasket. Residual gasket material or dirt can prevent the new gasket from sealing completely. An incorrectly sealed gasket may cause the engine to leak oil, lose compression, or overheat.

It is critical to completely remove old gasket material without damaging sealing surfaces. Aluminum is very soft and easy to scratch and gouge. A new gasket may not seal completely if the underlying surface is damaged. Generac does not recommend using a wire brush or abrasive sanding to remove gaskets. Use a metal gasket scraper specifically designed for this purpose. Scrape gasket at a shallow angle—almost parallel to the sealing surface. Wipe sealing surface with a clean, dry, lint-free rag after removing gasket residue.

If the gasket is difficult to scrape off, spray it with an aerosol gasket removal compound. Allow the compound to soak into the gasket for the length of time recommended by the manufacturer (typically 10– 20 minutes), then use the gasket scraper again.

NOTE: Some gasket removal compounds contain chemicals damaging to plastics. Read and follow the directions on the product label before use.

Assembling Cylinder Head

See [Assembling Cylinder Head](#) for information.

Section 10: Electric Starter

General Information

See [Figure 10-1](#). The starter motor utilizes a solenoid, similar to an automobile starter, to assist in pinion gear engagement. When the starter motor is activated, the pinion gear engages a ring gear attached to the engine flywheel and cranks the engine.

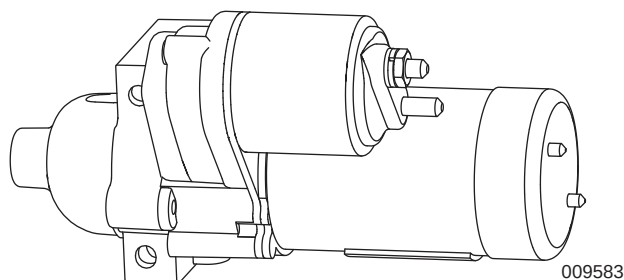


Figure 10-1. Starter Motor

Troubleshooting

IMPORTANT NOTE: Do not rotate engine with electric starter while spark plugs are removed, unless using a compression gauge. Arcing at the spark plug ends may ignite fuel vapor exiting the spark plug hole.

NOTE: If a starting problem is encountered, inspect the engine thoroughly to eliminate it as the cause of starting difficulty. Check engine for free rotation by removing spark plugs and flywheel guard and turning the crankshaft with a socket and ratchet or breaker bar. Verify it rotates freely.

Engine cranks slowly:

- Additional drag on engine, affecting starter performance (see note above).
- Discharged battery.
- Faulty electrical connection (battery circuit).
- Dirty or worn starter motor commutator, bearing, weak magnets, worn brushes, etc.
- Incorrect oil viscosity for temperature expected.
- Battery leads too long or wire too small.
- Underpowered battery.

Engine will not crank if:

- Faulty safety interlocks.
- Discharged battery.
- Faulty electrical connections.
- Faulty starter motor switch (open circuit).
- Open circuit in starter motor.

- Brushes sticking, etc.
- Faulty solenoid.

Starter motor spins but does not crank engine:

- Sticking pinion gear due to dirt.
- Damaged pinion or ring gear.
- Damaged or faulty battery.
- Incorrect rotation due to reversed motor polarity.
 - all motors rotate counterclockwise viewed from pinion gear.
- Damaged solenoid.

Starter motor spins but will not stop:

- Faulty starter switch.
- Solenoid stuck engaged.

Test Equipment

Digital Multimeter (DMM)

See [Figure 10-2](#). A DMM may be used to read volts, ohms, amperes, and test diodes (rectifiers).

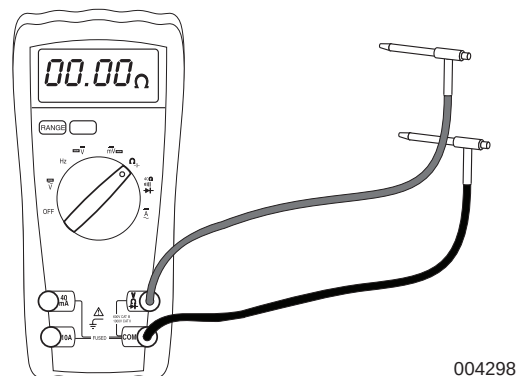


Figure 10-2. Digital Multimeter (DMM)

NOTE: The DMM is equipped with two fuses to prevent DMM damage if input limits are exceeded. Inspect fuses if the DMM displays a reading of 0.00 when testing DC output. See DMM operators' manual for more information.

Testing the Starter Motor

Testing Starter Solenoid

The solenoid is a normally open, electrically activated switch. Switch closes when key switch is set to START,

allowing battery current to flow to starter motor and crank engine.

NOTE: Solenoid may be tested while mounted on engine.

NOTE: A jumper test lead is required for this test.

Proceed as follows to test starter solenoid:

1. Remove positive (+) battery cable from battery.
2. Remove battery cable from stud terminal on solenoid.
3. Disconnect wire from tab terminal on solenoid.
4. Verify key switch is set to OFF.
5. Insert RED test lead into $V\Omega\rightarrow+$ receptacle on DMM.
6. Insert BLACK test lead into **COM** receptacle on DMM.
7. Set DMM selector to $\Omega\rightarrow+$ .
8. See [Figure 10-3](#). Attach one DMM test lead to each stud terminal on starter solenoid.

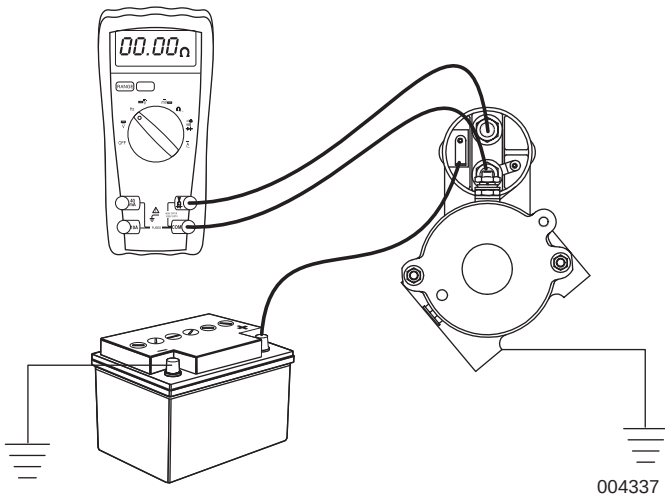


Figure 10-3. Test Starter Solenoid

9. Attach one end of jumper lead to positive terminal on battery.
10. Touch jumper wire to tab terminal on solenoid. A “click” should be heard as the solenoid switch closes. DMM should make a continuous tone and show continuity.

Verifying Starter Voltage Drop

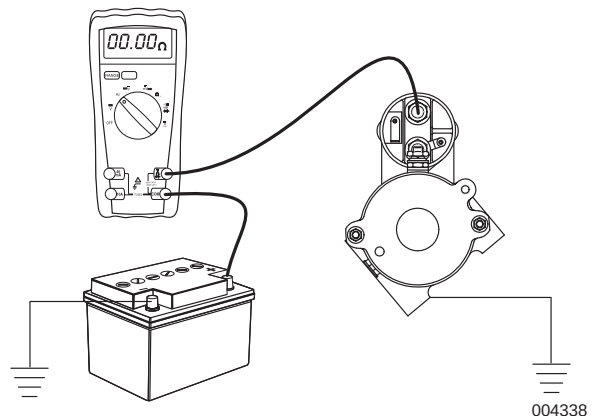
Verify battery is at least 70% charged (12.5 V or higher) before starting this test. Verify a correctly sized battery is being used. See engine manufacturer's operator's manual for battery size recommendations.

Measure battery voltage across the battery posts using a DMM set to the 20V scale with engine and all other accessories turned off. Do not measure across battery wire terminal ends. Voltage should read 12.5 V or higher.

NOTE: Do not crank engine longer than 15 seconds during any of the following tests.

Proceed as follows to verify starter voltage drop:

1. To prevent the engine from starting: disconnect the ECM, manually open throttle plate on the ETC, and connect a remote button starter switch from the battery positive (+) to the starter relay.
2. Place DMM positive (+) lead on positive battery terminal and negative (-) lead on negative battery terminal. Crank engine and observe DMM reading.
3. Place DMM positive (+) lead on starter battery terminal stud and negative (-) lead on starter housing. Crank engine and observe DMM reading.
4. Compare the two voltage readings. If readings are the same (or vary within one volt or less), there is no excessive voltage drop on the positive side. If a voltage drop of greater than one volt exists, proceed to step 5.
5. See [Figure 10-4](#). Set DMM on the DC 2V scale. Place DMM negative (-) lead on positive (+) battery terminal and DMM positive (+) lead on starter positive (+) terminal stud. With engine still disabled, crank engine and observe DMM reading. There should be less than a 0.2 V drop. If drop is greater than 0.2 V, test each wire and terminal connection. No more than 0.2 volts should be observed at each connection.



6. Test negative (-) side of system. Place DMM positive (+) lead on starter case and DMM negative (-) lead on negative (-) battery terminal.
7. Crank engine and observe the reading. Voltage should be less than 0.2 V.
8. If voltage reads greater than 0.2 volts, test all wire connections on the starter negative (-) side. There should be no readings higher than 0.2 volts.

Testing Starter Amp Draw

CAUTION

Equipment Damage. Do not continuously crank engine for more than ten seconds. Doing so will lead to overdischarge of batteries and starter seizure. (C000230)

This test requires a special clamp-on tool called an ammeter. An engine repair shop should be able to perform this test.

Proceed as follows to disable engine from starting for testing:

1. Disconnect ECM.
2. Manually open throttle plate on the ETC.
3. Connect a remote button starter switch from the battery positive (+) to the starter relay.

Proceed as follows to test starter amp draw:

1. Place DC ammeter clamp on positive (+) cable running from the positive (+) battery terminal to starter post.
2. Ground ignition and crank engine for no longer than 10 seconds. Starter amp draw should be less than 100 amps during cranking.

Excessive draw could be caused by the following:

- Faulty starter. To verify, remove starter from engine and bench test prior to replacing. Perform an amp draw test on the bench. Starter should not exceed 62 amps with a minimum pinion speed of 3,250 rpm.
- Engine oil viscosity too heavy (see engine manufacturer's operators manual).
- Some equipment manufacturers may use a constant drive pump. Constant drive pump oil may be too heavy or belt tension may be too tight.
- Internal engine problem. Turn engine over with a socket and ratchet to verify it rotates easily.
- Battery cable loose or incorrectly sized.
- Battery terminals corroded.
- Battery CCA too low.

Battery Information



WARNING

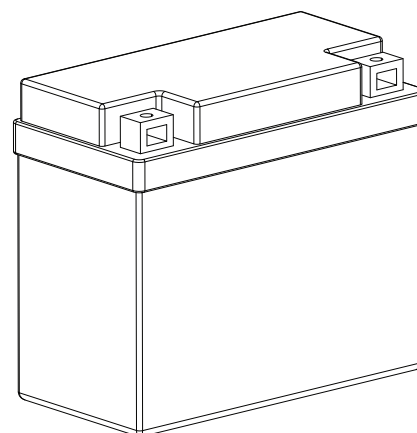
Explosion. Batteries emit explosive gases. Always disconnect negative battery cable first to avoid spark. Failure to do so could result in death or serious injury. (W000238)



WARNING

Vision loss. Eye protection is required when servicing unit. Failure to do so could result in vision loss or serious injury. (W000377)

A 12 volt battery is required to operate the starter motor on Generac G-Force V-Twin engines. Battery rating should meet the minimum battery recommendations for best starter life and performance. See unit owner's manual for correct battery requirements.



020592

Figure 10-4. Typical Battery

Battery Cable Recommendations:

These cable sizes are based on total length of cable from positive (+) battery terminal to starter, plus ground return to negative (-) battery terminal.

- #4 AWG — 6 ft (1.8 m) or less
- #2 AWG — 12 ft (3.7 m) or less

Checking Battery

Proceed as follows to check battery:

1. Inspect and clean if necessary. Inspect for:
 - corrosion
 - dirt
 - terminals and clamps—good condition
 - terminals and clamps—secure connections
2. Fully charge battery.

IMPORTANT NOTE: DO NOT exceed charge rate of 1/10 ampere for every ampere of battery rating. Contact battery manufacturer for maximum charge recommendations. Use a trickle charger to reduce charge rate.

NOTE: Periodically unplug charger if battery becomes hot to the touch or emits excessive gas or electrolyte solution.

Testing Battery



Equipment Damage. Do not continuously crank engine for more than ten seconds. Doing so will lead to overdischarge of batteries and starter seizure.

(C000230)

Proceed as follows to test battery:

1. Set DMM to read DC Volts.
2. Attach red DMM test lead to positive (+) battery terminal. Attach black DMM test lead to negative (-) battery terminal.
3. Set ignition switch to OFF.
4. Press starter button. If ignition switch and starter switch are the same switch, disconnect wires from spark plugs and ground ignition using two ignition testers.
5. Set ignition switch to START. DMM should display 9 volts or more while cranking engine. Replace battery if less than 9 volts is measured.

Battery Disposal

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit Call2Recycle website at: [http://
Call2Recycle.org/locator](http://Call2Recycle.org/locator).

Section 11: Lubrication System

Description

The Generac G-Force V-Twin engines uses a full pressure lubrication system with an oil filter. A gerotor oil pump draws oil from a screened oil pickup in the sump and pumps oil through the oil filter.

Filtered oil flows through an oil gallery in the sump and is distributed to the main bearings, connecting rod bearings, hydraulic lash adjusters, and camshaft bearings. Engine oil pressure varies with oil viscosity, ambient air temperature differences, operating temperatures, and engine load.

- Oil Pressure - @ 70 °F (21 °C): 15–60 psi (103–414 kPa)

Protection Systems

Low Oil Pressure Switch

See [Figure 11-1](#). The engine is usually equipped with a normally open type, low oil pressure switch (A) which closes the circuit between the terminals when oil pressure drops below 10 +/- 3 psi (69 kPa). The engine will shut down on low oil pressure if one terminal is connected to the ignition ground wire and the other terminal is connected to ground. The low oil pressure switch can also be connected to an alarm or light to indicate low oil pressure, instead of shutting down the engine.

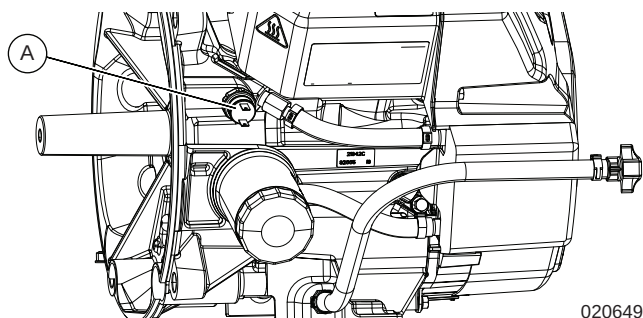


Figure 11-1. Low Oil Pressure Switch

A delay built into the shutdown system on some engine applications allows oil pressure to build during starting. The delay allows the engine to run for about 10 seconds before sensing oil pressure. The engine shuts down if the engine is wired for low oil pressure shutdown and the system senses low oil pressure during operation. The engine will not run until 8 psi (55 kPa) of oil pressure is reached. If a restart is attempted within five seconds after shutdown, the engine may NOT start. The system needs 10 seconds to reset.

Checking Engine Oil Level



Engine damage. Verify proper type and quantity of engine oil prior to starting engine. Failure to do so could result in engine damage.

(C000135)

Oil capacity of the Generac G-Force V-Twin engines is approximately 2.2 qt (2.1L). See [Specifications](#) for the recommended amount for each model.

Proceed as follows to check engine oil level:

1. See [Figure 11-2](#). Remove oil dipstick (A) and wipe it dry with a clean cloth.

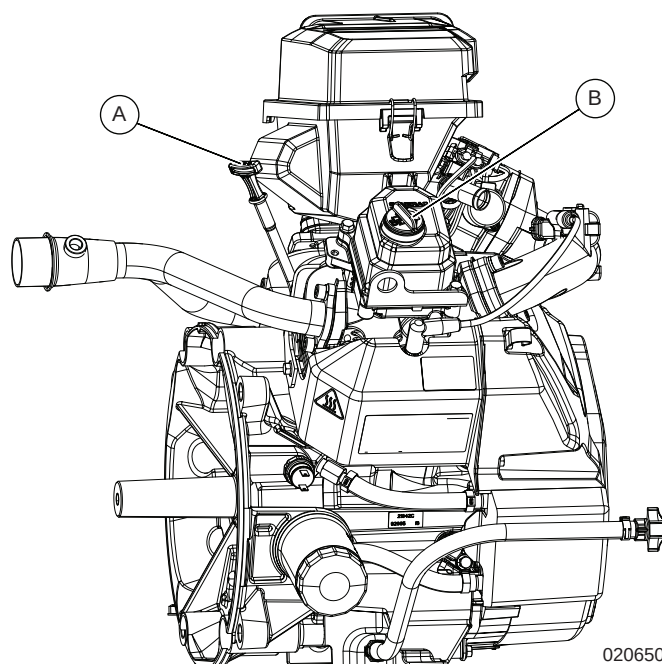


Figure 11-2. Oil Fill/Oil Dipstick Location

2. See [Figure 11-3](#). Insert oil dipstick completely, then remove. Oil level (C) should be at the FULL mark.

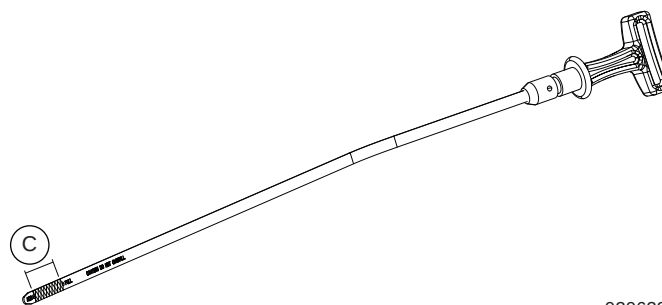


Figure 11-3. Oil Dip Stick Fill Indication

NOTE: Never operate engine with oil level below the ADD mark on the oil dipstick. Doing so could damage the engine.

3. See [Figure 11-2](#). If necessary, add oil through location in rocker cover until the FULL mark is reached. Do not fill above the FULL mark.

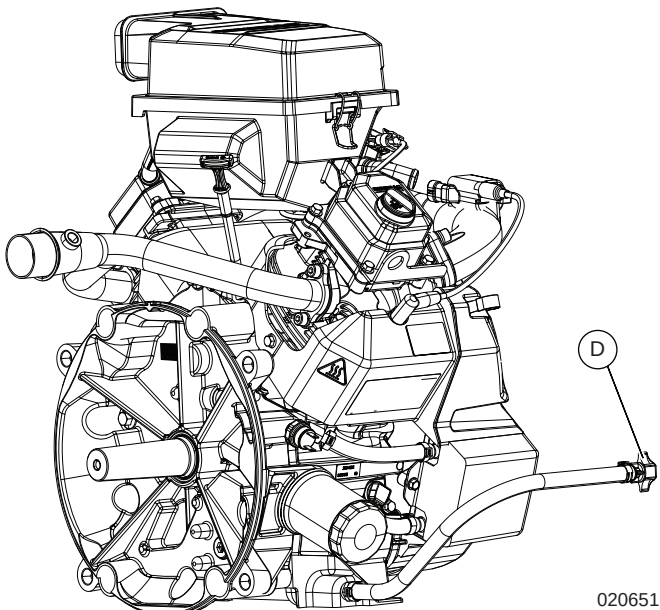
Changing Engine Oil And Filter

See [Specifications](#) for crankcase oil capacities. Do not use special additives. Verify unit is level when filling with oil. DO NOT OVERFILL.

The engine is equipped with an oil filter. Change oil and oil filter after the first eight hours of operation. Change oil and oil filter every 100 hours thereafter. In dirty or dusty operating conditions, or in extremely hot weather, change oil more often.

Proceed as follows to change oil while engine is still warm:

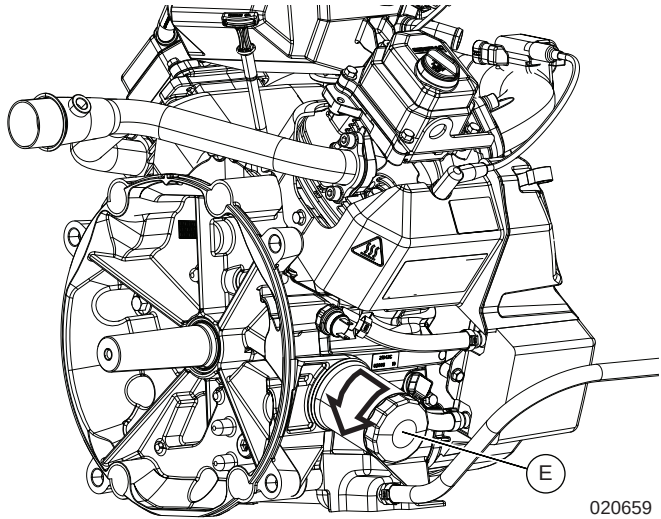
1. See [Figure 11-4](#). Clean area around oil drain plug (D). Remove oil drain plug and drain oil completely into a suitable container.



020651

Figure 11-4. Oil Drain Plug

2. Install and tighten oil drain plug after oil is completely drained.
3. See [Figure 11-5](#). To change oil filter:



020659

Figure 11-5. Remove Oil Filter

- a. Locate oil filter (E).
- b. Place a suitable container beneath oil filter and rotate filter counterclockwise to remove.
- c. Coat gasket on new oil filter with clean engine oil. Install and rotate new oil filter clockwise until gasket contacts oil filter adapter, then tighten an additional three-quarters turn.
4. See [Figure 11-2](#) and [Figure 11-3](#). Remove oil fill cap and insert a clean funnel into oil fill opening (B). Fill crankcase with recommended oil until oil level is at FULL mark (C) on oil dipstick (A).
5. Install oil fill cap after crankcase is filled to correct level. Start engine to fill oil filter. Verify and adjust oil level as needed.

Section 12: Engine Assembly

Long Block Assembly

Installing Gear Cover

Proceed as follows to install oil seal and check torque on gear cover assembly:

1. See [Figure 12-1](#). Press crankshaft PTO oil seal (A) into gear cover assembly (B) until seal is fully seated against bottom of bore.

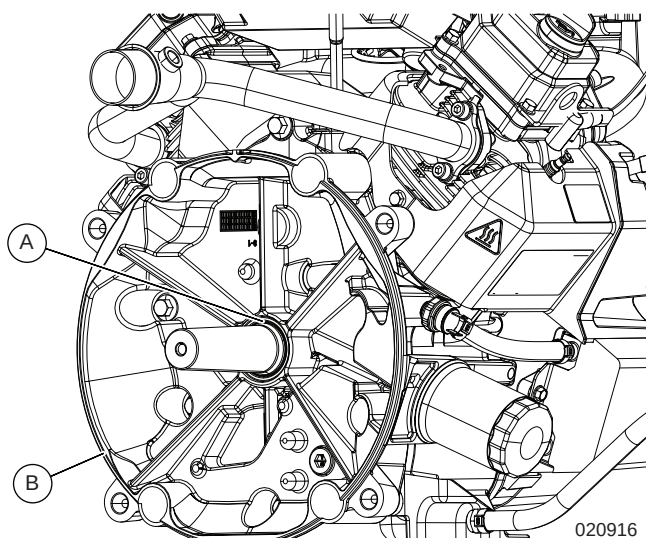


Figure 12-1. Install Oil Seal Into Gear Cover

2. See [Figure 12-2](#). Verify gear cover torque using the correct sequence shown. Final torque on gear cover is 32.4–39.6 ft-lb (43.9–53.7 Nm).

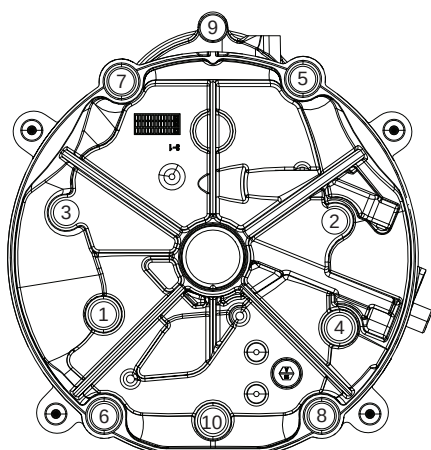


Figure 12-2. Gear Cover Torque Sequence

Installing Cylinder Heads

Proceed as follows to install cylinder heads:

1. See [Figure 12-3](#). Install cylinder head bolt dowels (E) into cylinder head assembly.

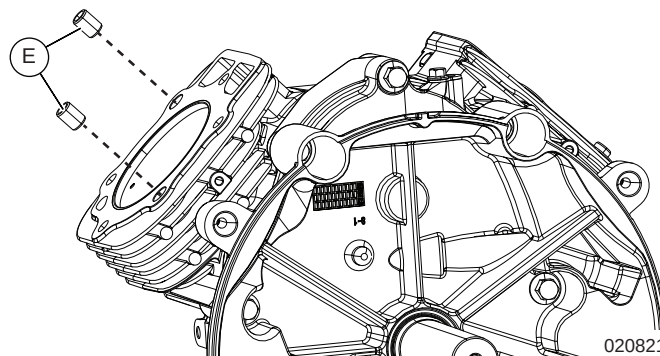


Figure 12-3. Install Cylinder Head Dowels into Crankcase

2. See [Figure 12-4](#). Install cylinder head gasket (D) on engine long block. Install cylinder head (C) on long block assembly.

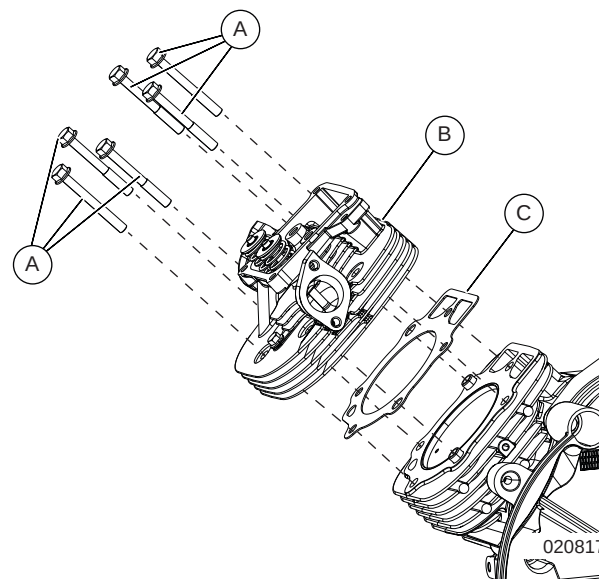
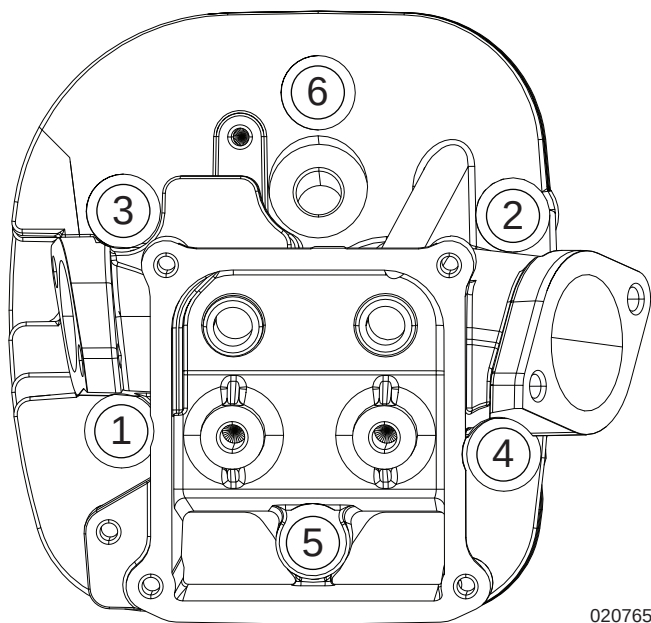


Figure 12-4. Install Cylinder Head onto Crankcase

3. Install five cylinder head bolts (A) and washers (B).
4. See [Figure 12-5](#). Tighten cylinder head bolts in pattern shown with the two step process as follows.
 - a. First sequence, tighten using pattern to 9–11 ft-lb (13–15 Nm).
 - b. Final sequence, tighten using pattern to 19.5–22.4 ft-lb (26.5–30.4 Nm).



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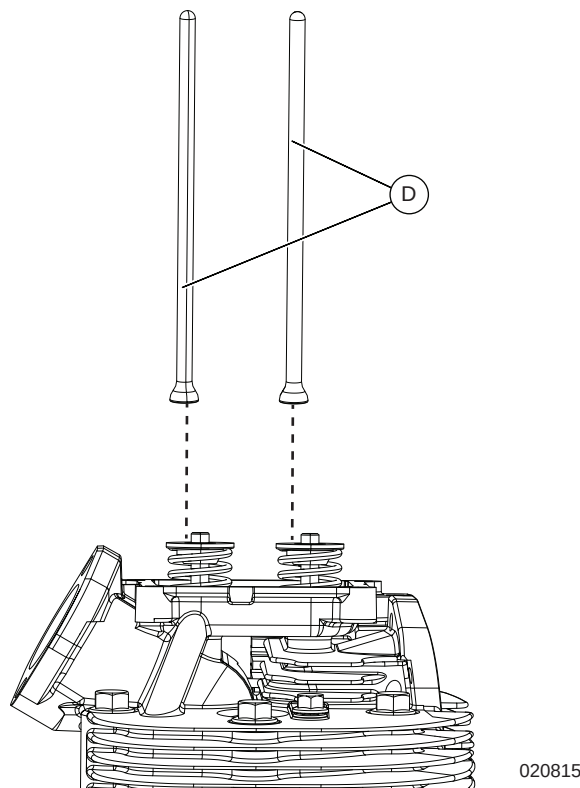
Figure 12-5. Cylinder Head Bolt Torque Sequence

5. Repeat cylinder head assembly steps 1– 3 for second cylinder head using the same components stated above.

Installing Valve Train

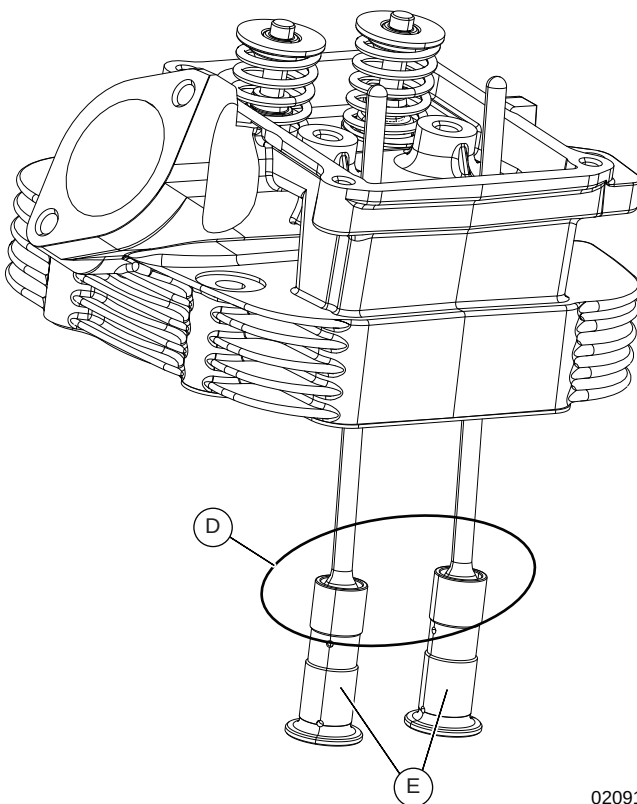
Proceed as follows to install valve train:

1. See [Figure 12-6](#) and [Figure 12-7](#). Apply assembly lube to each end of two pushrods (D). Install pushrods into cylinder one. Verify each pushrod is correctly engaged in the recesses of the tappets (E).



020815

Figure 12-6. Install Pushrods into Cylinder Head



020918

Figure 12-7. Install Pushrods into Tappets

2. See [Figure 12-8](#). Install pushrod guide plate (C), rocker arms (B), and rocker arm pivot studs (A) loosely onto cylinder head. Verify ball end of

pushrod tubes are inserted into ball recess in rocker arms.

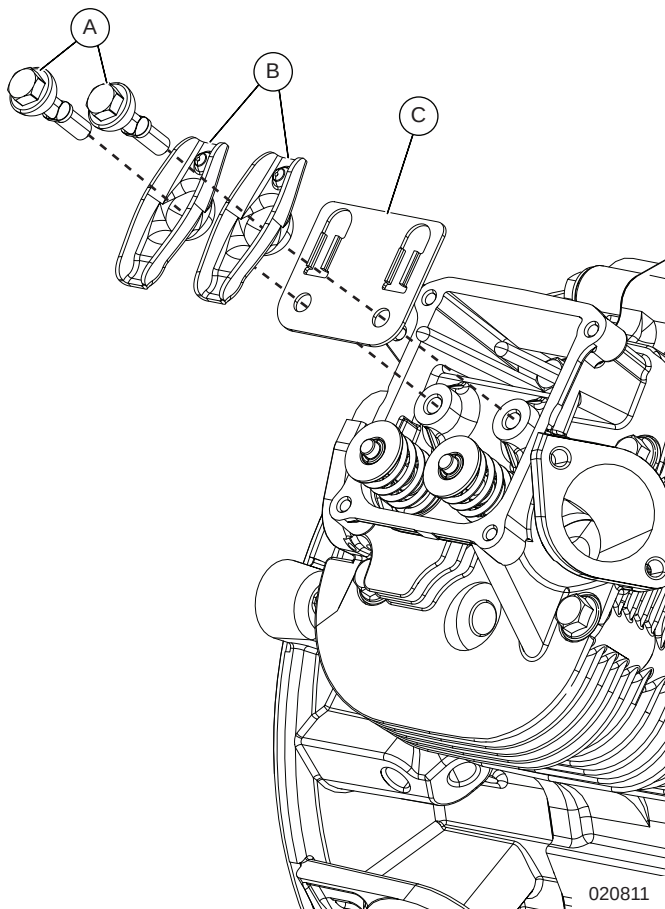


Figure 12-8. Install Rocker Arm into Cylinder Head

3. Rotate crankshaft clockwise, as viewed from flywheel side, until piston is at TDC of compression stroke (intake valve just closed).
4. Spin pushrod using fingers while tightening pivot stud until resistance is felt on pushrod.
5. Tighten rocker arm pivot studs to 11.1-12.4 ft-lbs (15.1-16.8 Nm).
6. Repeat steps 3–5 for exhaust valve.
7. Repeat cylinder head assembly steps 1–6 for second cylinder head using the same components stated above.

Installing Rocker Cover

Proceed as follows to install rocker cover:

1. See [Figure 12-9](#). Assemble coalescing media (A), breather gasket (B), breather plate (C), and fasteners (D) into rocker cover (E) for cylinder 2 to build breather assembly.

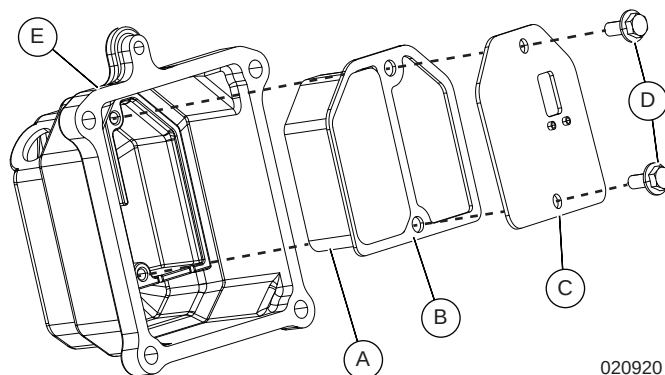


Figure 12-9. Install Oil Breather System into Rocker Cover

2. See [Figure 12-10](#). Install rocker cover gasket (J) onto cylinder 1. Use rocker cover bolts (F) to install rocker cover (H) and lifting bracket (G).

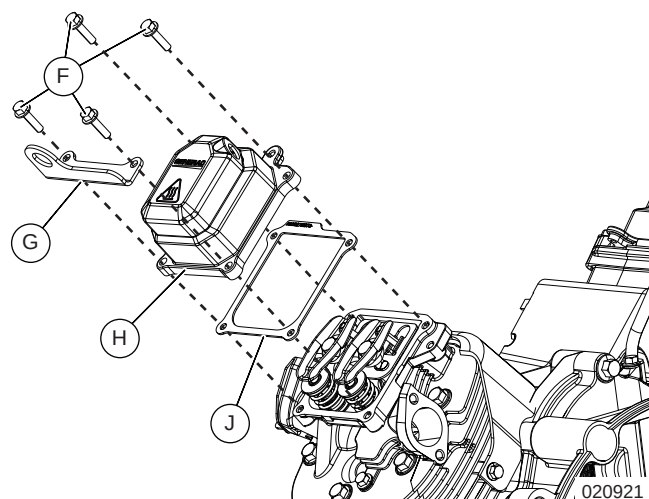


Figure 12-10. Install Rocker Assembly onto Cylinder Head

3. Tighten rocker cover assembly to 58–66 in-lb (6.5–7.5 Nm).
4. Repeat steps 1–2 using rocker cover with breather hole and components stated therein onto cylinder 2. Install lifting bracket on PTO side of engine.
5. See [Figure 12-11](#). Install breather hose (K) into rocker cover facing PTO side of engine. Soapy water or spray silicone lubricant may be needed to fit breather hose into rocker cover hole.

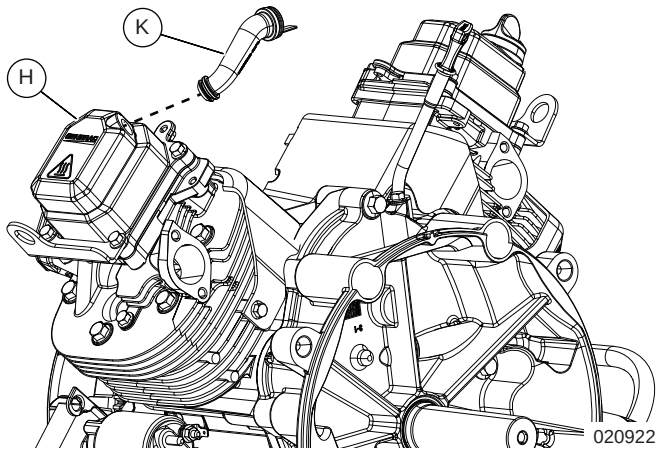


Figure 12-11. Install Breather Hose into Rocker Cover

Installing Head Temperature Sensor

Proceed as follows to install head temperature sensor:

1. See [Figure 12-12](#). Install head temperature sensor (A) with fastener (B).

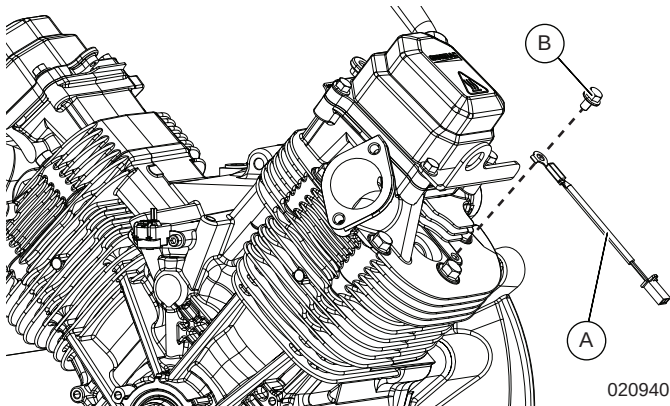


Figure 12-12. Install Head Temperature Sensor

2. Tighten temperature fastener to 57.5–66.3 in-lb (6.5–7.5 Nm).

Installing Backing Plate

Proceed as follows to install backing plate:

1. See [Figure 12-13](#). Install backing plate (A) onto PTO side of engine.

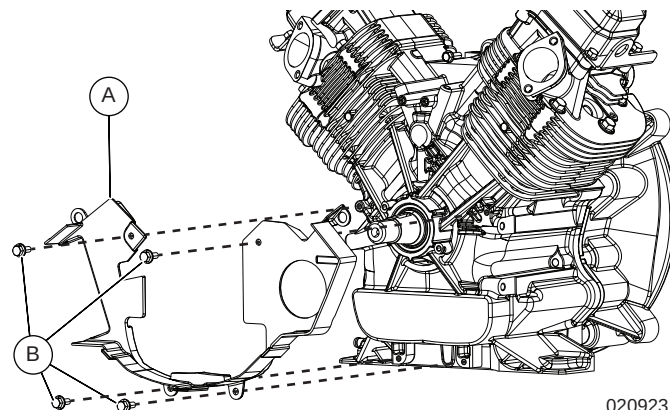


Figure 12-13. Install Backing Plate onto Crankcase

2. Tighten fasteners (B) to 70.7–88.7 in-lbs (7.99–10.02 Nm).

Installing Ground Wire and Stator

Proceed as follows to install ground wire and stator:

1. Install ground wire connector of ignition coil onto backing plate.
2. See [Figure 12-14](#). Push to install clamp wire retainer (A) to crankcase.

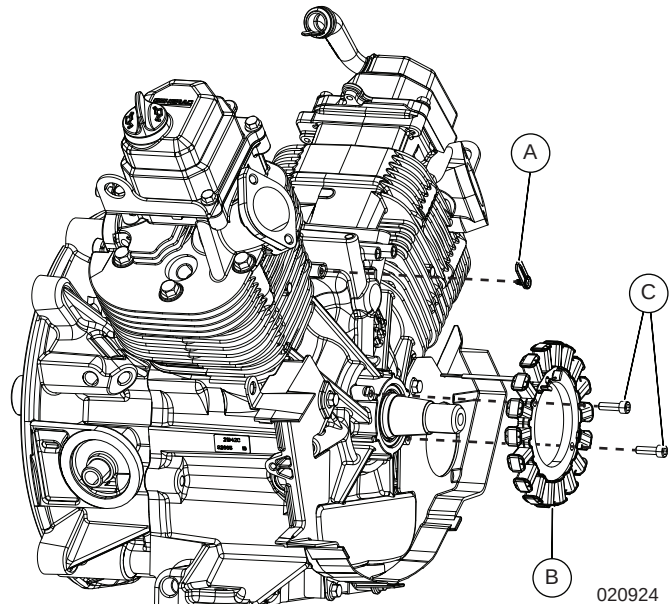


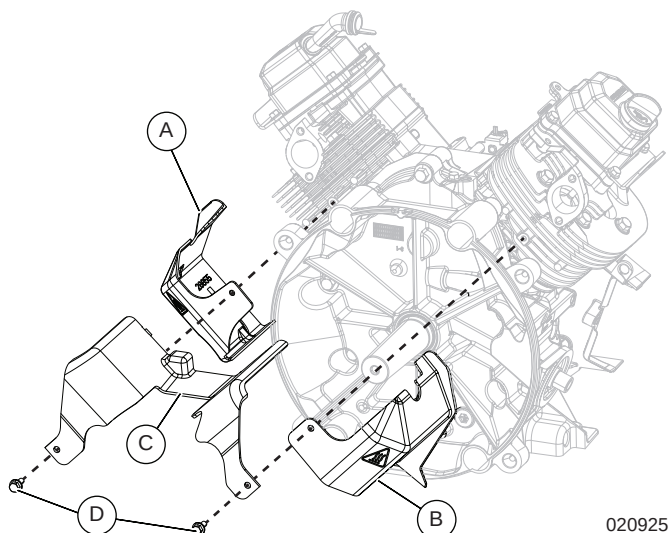
Figure 12-14. Install 1-Phase Stator

3. Install 1-phase stator (C) using two fasteners (D).
4. Tighten stator fasteners to 97–120 in-lb (11–13.6 Nm).

Installing Cylinder Wrappers

Proceed as follows to install cylinder wrappers:

1. See [Figure 12-15](#). Install cylinder wrappers (A, B, C) to crankcase using two fasteners (D) (one fastener per cylinder). Top wrapper (C) is installed on top of side wrappers (A, B).



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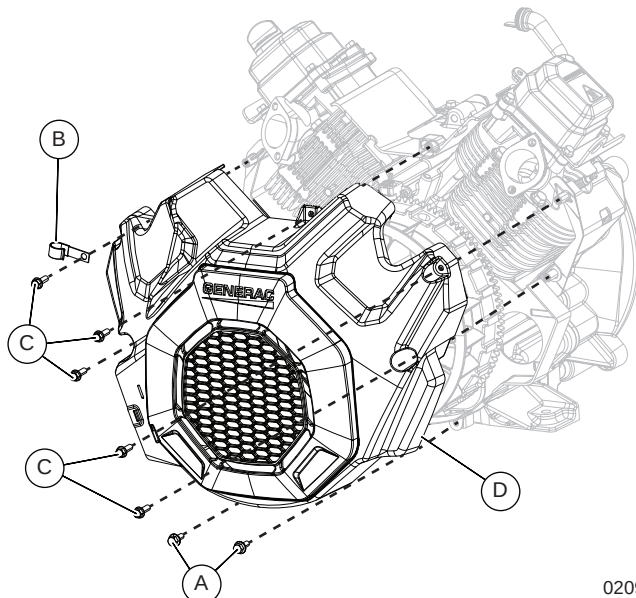
Figure 12-15. Install Cylinder Wrappers

2. Tighten fastener to 70.7 - 88.7 in-lbs (7.99-10.02 Nm).

Installing Blower Housing

Proceed as follows to install blower housing:

1. See [Figure 12-16](#). Remove two bottom backing plate fasteners (A) (retain for later use).



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Figure 12-16. Install Blower Housing

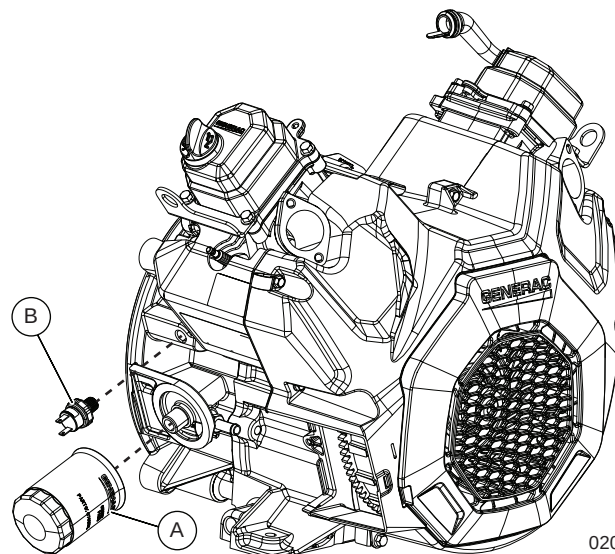
2. Install blower housing (D) by fastening it to cylinder wrappers using five fasteners (C).
3. Install J-clip (B) to area noted using appropriate blower housing fastener.
4. Install backing plate fasteners removed in first step.
5. Tighten fasteners to 10 in-lb (1.13 Nm).

Installing External Oil Components

Installing Oil Filter and Pressure Switch

Proceed as follows to install oil filter and pressure switch:

1. See [Figure 12-17](#). Apply small amount of clean engine oil to gasket on oil filter (A).



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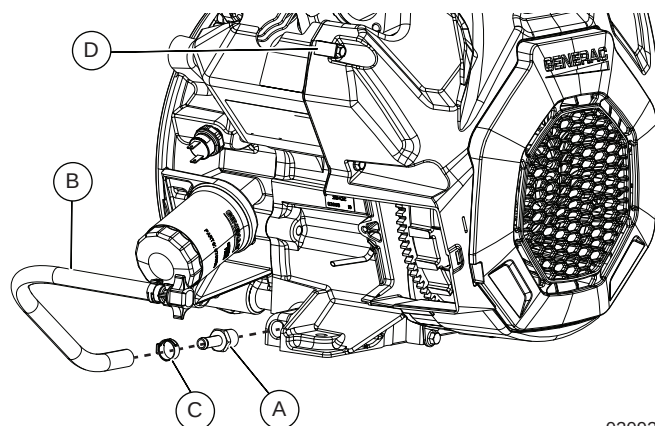
Figure 12-17. Install Oil Filter and Oil Pressure Switch

2. Screw onto threaded oil filter mount.
3. Tighten oil filter to 84 in-lb (9.5 Nm).
4. Verify oil pressure switch (B) has lock patch on threads.
5. Install switch in gear cover below oil filter.
6. Tighten switch two to three turns from finger tight.

Installing Oil Drain and Oil Dipstick

Proceed as follows to install oil drain line and oil dipstick:

1. See [Figure 12-18](#). Verify 3/8 -18 NPT* hose barb oil drain fitting (A) has lock patch on threads.



020928

Figure 12-18. Install Oil Drain Assembly

2. Install hose barb oil drain fitting into crankcase.

NOTE: *NPT Fittings: As a general rule, pipe fittings with tapered threads should not be assembled to a specific torque. Instead, they should be installed by the number of turns past finger tight. All NPT plugs below 1 in size should be installed to 2-3 turns from finger tight (T.F.F.T.).

3. Fit oil drain assembly over barbed fitting and secure with 16.5 mm stepless ear clamp (C).
4. Hang oil drain hose in the J-clip (D) on blower housing.
5. See [Figure 12-19](#). Fill engine through port in rocker cover (F) with 2.5 qt (2.4 L) of clean Generac 5W-20 synthetic engine oil (dry oil filter).

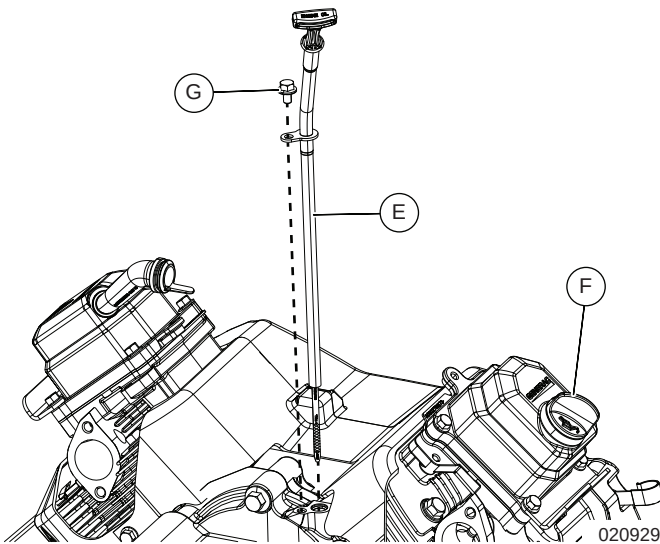


Figure 12-19. Install Dipstick

6. Install dipstick assembly (E) in port at the top of engine.
7. Secure mount with fastener (G).
8. Tighten fastener to 97.3 - 115.1 **in-lbs** (11-13 Nm).

Installing Oil Level Switch

Proceed as follows to install oil level switch:

1. See [Figure 12-20](#). Carefully rotate and guide oil level switch (B) into port on side of engine.

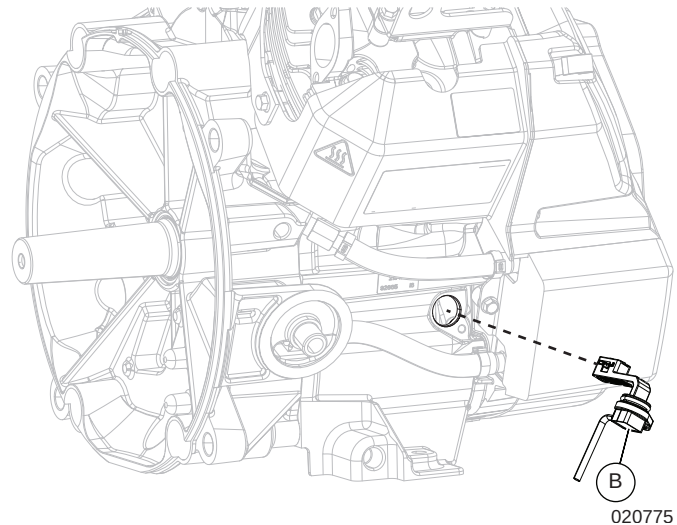
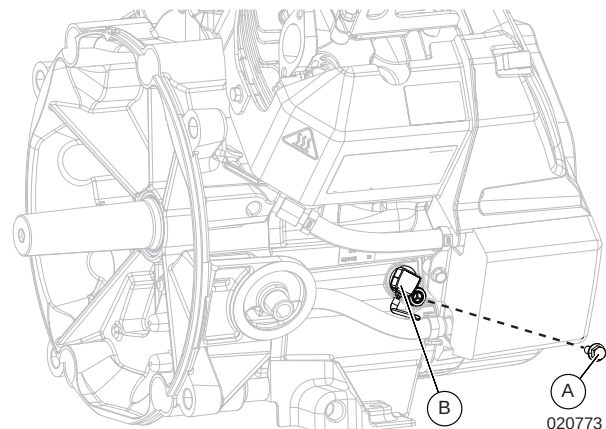


Figure 12-20. Install Switch

2. See [Figure 12-21](#). Install screw (A) fastening the oil level switch (B) to engine.



Installing Oil Cooler

Proceed as follows to install oil cooler:

1. See [Figure 12-22](#). Verify 1/4 NPT* x 3/8 hose barb oil cooler fitting (A) has lock patch on threads.

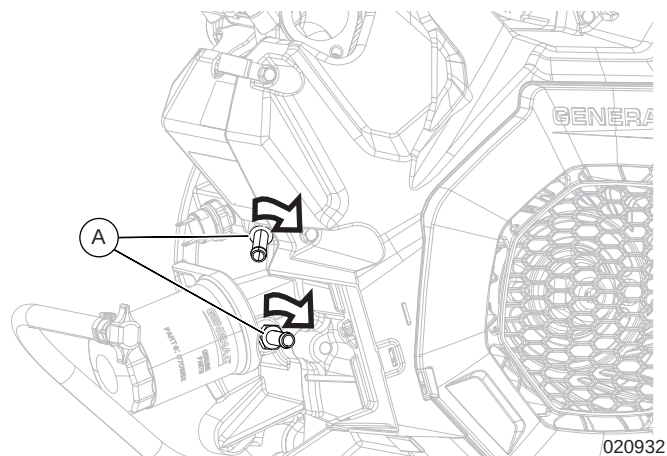


Figure 12-21. Oil Cooler Barbed Fittings

NOTE: *NPT Fittings: As a general rule, pipe fittings with tapered threads should not be assembled to a specific torque. Instead, they should be installed by the number of turns past finger tight. All NPT plugs below 1 in size should be installed to 2-3 turns from finger tight (T.F.F.T.).

2. Install hose barb oil cooler fittings on engine.
3. See [Figure 12-23](#). Insert blower side of oil cooler into blower housing. Verify oil cooler tabs (C) are inserted in oil cooler mounting slots (B).

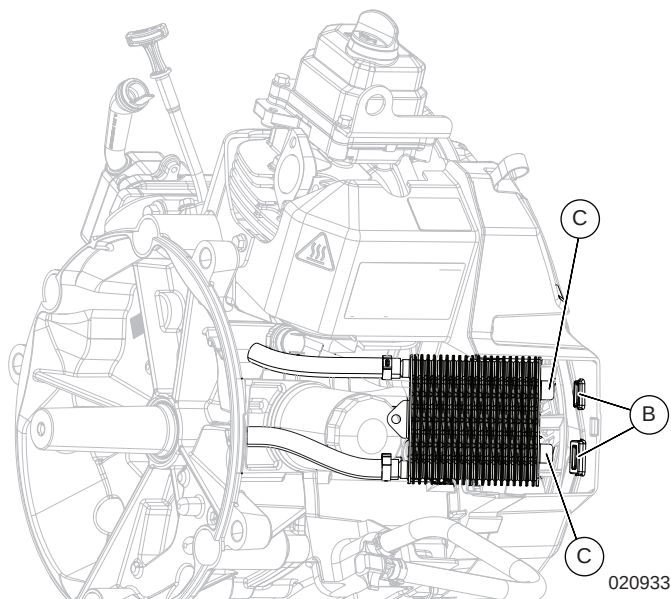


Figure 12-22. Install Oil Cooler

4. See [Figure 12-24](#). Install rubber cooler hoses (D) onto engine barbed fittings (A).

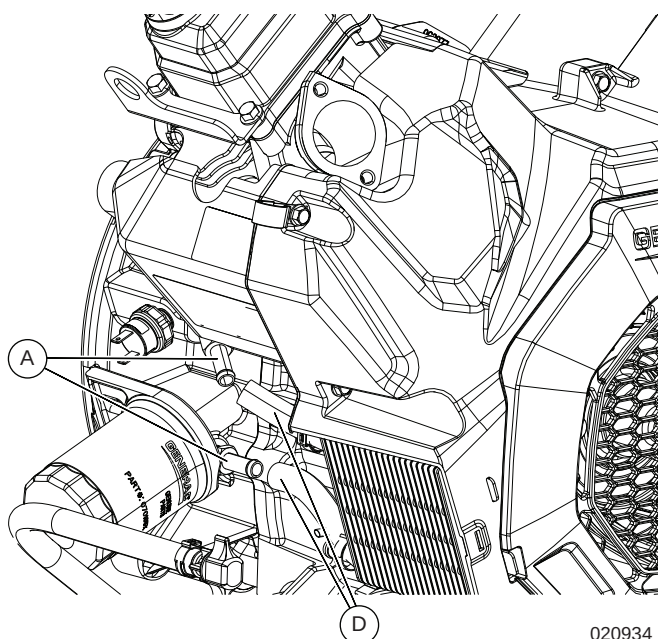


Figure 12-23. Install Rubber Hoses

5. See [Figure 12-25](#). Install new stepless ear clamps (E) to secure hoses to barbed fittings.

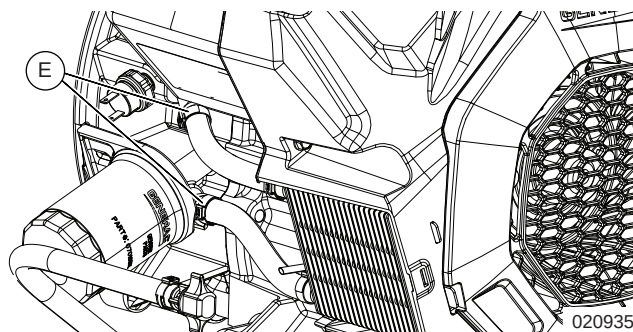


Figure 12-24. Install Stepless Ear Clamps

6. See [Figure 12-26](#). Install oil cooler cover (F) over oil cooler and secure with fastener (G).

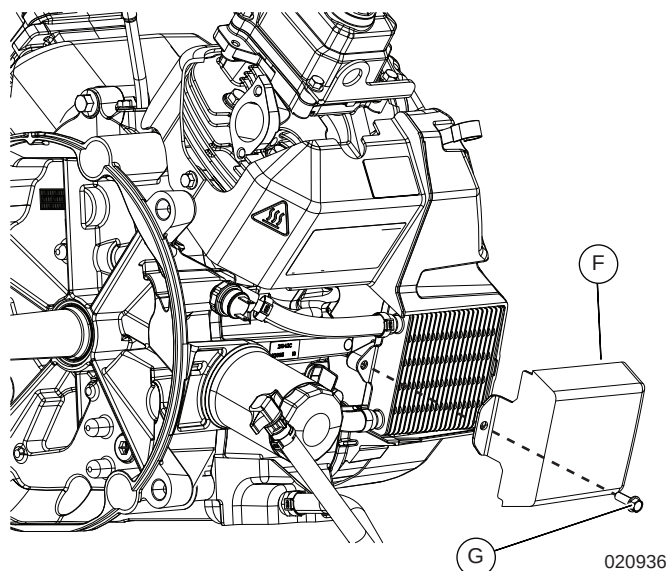


Figure 12-25. Install Cooler Cover

Installing Spark Plugs

Proceed as follows to install spark plugs:

1. See [Figure 12-27](#). Install spark plugs (A) into both cylinders.

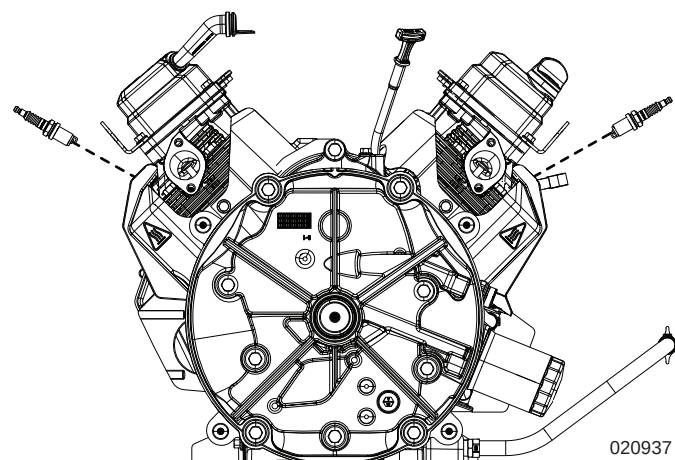


Figure 12-26. Install Spark Plugs

2. Tighten spark plugs to 14.8–22.1 ft-lb (20–30 Nm).

Installing Intake and Exhaust

Installing Intake Manifold

Proceed as follows to install intake manifold:

1. See [Figure 12-28](#). Place two intake manifold gaskets (C) onto cylinder heads (one on each intake port).

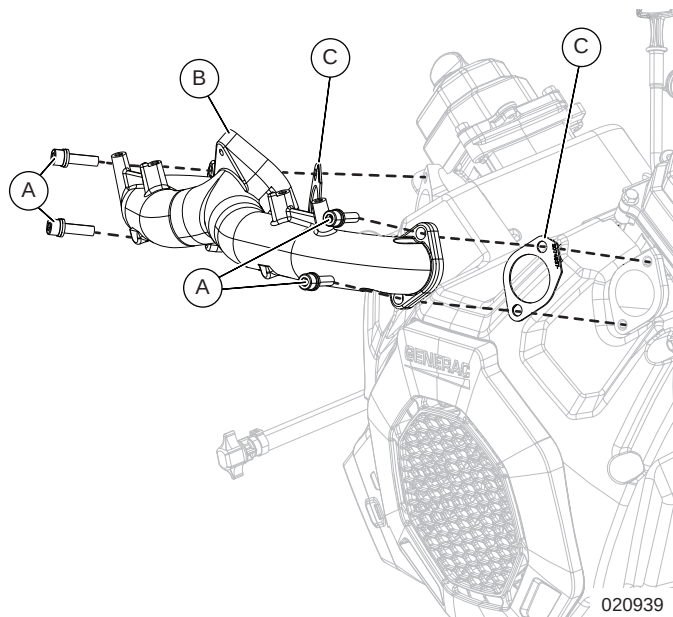


Figure 12-27. Install Intake Manifold onto Cylinder Head

2. Install intake manifold (B) onto cylinder heads using four fasteners (A).
3. Tighten fasteners to 14–16 ft-lb (19–22 Nm).

Installing Throttle Body and Air Box

Proceed as follows to install throttle body and air box:

1. See [Figure 12-29](#). Install throttle body assembly (A) and air box assembly (B) onto intake manifold using two bolts (C). Verify air box bolt holes align with the tabs on the rocker covers.

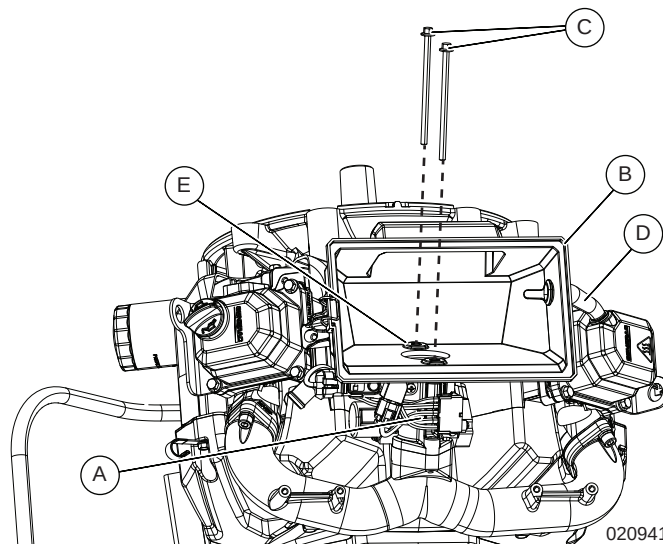


Figure 12-28. Install Air Box

2. Tighten bolts to 7.4–8.5 ft-lb (10–11.5 Nm).
3. Install breather hose (D) into mount in air box. Take care not to damage the hose on insertion into the hole on the air box. Soapy water or silicone spray may be needed to assist the insertion.
4. Install bolts (E) through holes and grommets on tappet cover mounts.

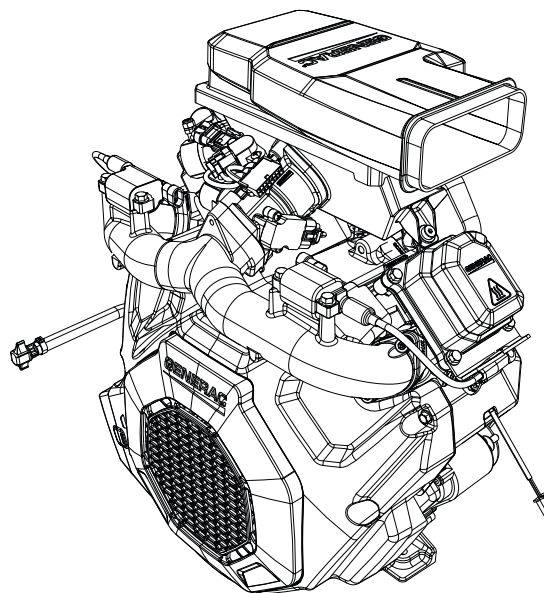


Figure 12-29. Completed Installation

Section 13: Troubleshooting

PROBLEM	POSSIBLE CAUSES	REMEDY
Engine does not start	Low oil level.	Replenish oil as necessary.
	Dirty air cleaner.	Clean or replace air cleaner element.
	Obstructed fuel line.	Clean or replace fuel line.
	Fouled spark plug.	Replace spark plug.
	Incorrect spark plug.	Replace with correct spark plug.
	Loose or faulty ignition wiring.	Inspect wiring; repair or replace wiring as necessary.
	No ignition spark.	Inspect ignition system; replace faulty part(s).
	Incorrect valve clearance.	Inspect pushrods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Poor compression.	Inspect for worn or scored cylinder, possible burnt valve, or burnt valve seat.
	Insufficient fuel.	Replenish fuel as necessary.
Engine knocks	Carbon in combustion chamber.	Clean carbon from head and piston.
	Loose flywheel.	Inspect flywheel key and keyway; replace parts if necessary. Tighten flywheel nut.
	Worn cylinder.	Replace crankcase.
	Loose or worn connecting rod.	Replace connecting rod. Verify crankpin diameter to see if crankshaft is faulty.
	Incorrect valve clearance.	Inspect pushrods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Overloaded engine.	Reduce excessive load.
Engine misses under load	Incorrect spark plug(s).	Install correct spark plug(s).
	Fouled spark plug(s).	Replace spark plug(s).
	Cracked porcelain on spark plug(s).	
	Incorrect spark plug gap.	Adjust spark plug gap.
	Incorrect valve clearance.	Inspect pushrods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Weak valve spring(s).	Replace weak valve spring(s).
Engine lacks power	Incorrect spark plug(s).	Install correct spark plug(s).
	Incorrect valve clearance.	Inspect pushrods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Worn piston rings.	Replace piston rings.
	Low oil level.	Replenish oil as necessary.
	Obstructed air cleaner.	Clean or replace air cleaner.
	Worn or burned valves or valve seats.	Grind valves and valve seats.

PROBLEM	POSSIBLE CAUSES	REMEDY
Engine overheats	Obstructed air flow.	Remove obstructions.
	Clogged cooling fins.	Clean cooling fins.
	Carbon buildup in combustion chamber.	Remove cylinder head and clean carbon.
	Overloaded engine.	Reduce excessive loading.
	Insufficient lubrication.	Replenish oil as necessary.
Engine vibrates excessively	Loose engine mounting bolts.	Tighten mounting bolts.
	Bent crankshaft.	Replace crankshaft.
	Problem in equipment connected to engine.	Inspect equipment driven by engine.
Engine surges or runs unevenly	Intermittent ignition spark.	Test ignition as per Ignition ; repair or replace faulty parts as necessary.
Engine uses excessive amounts of oil	Engine speed too fast.	Adjust engine speed to specifications.
	Oil level too high.	Drain excess oil from crankcase.
	Loose oil filler cap or damaged oil filler cap gasket.	Replace cap gasket and tighten.
	Damaged or dirty breather.	Clean or replace breather as necessary.
	Clogged drain holes in breather.	Unclog drain holes in breather.
	Leaking gaskets or seals.	Replace gaskets or seals.
	Excessively worn valve guides.	Replace cylinder head.
	Worn cylinder walls.	Replace crankcase.
	Worn piston rings and/or grooves.	Install new rings and/or piston.
	Obstructed oil passages.	Clean out oil passages.
Crankshaft oil seal leaks	Dirty or faulty crankcase breather.	Clean or replace breather.
	Hardened or worn oil seal.	Replace oil seal.
	Worn crankshaft seal contact surface.	Verify crankshaft size and replace if worn.
	Worn crankshaft sleeve bearing.	Replace sleeve bearing.
	Seal not seated squarely in bore.	Replace with new seal.
	New seal damaged during installation.	
	Bent crankshaft.	Inspect crankshaft for straightness; replace if necessary.
Breather is passing oil	Oil level too high.	Drain excess oil from crankcase.
	Loose oil filler cap or damaged oil filler cap gasket.	Replace cap gasket and tighten cap.
	Damaged or dirty breather.	Clean or replace breather as necessary.
	Clogged drain holes in breather.	Unclog drain holes in breather.
	Aligned piston ring end gaps.	Rotate ring end gaps, stagger 90°.

PROBLEM	POSSIBLE CAUSES	REMEDY
	Rings seated incorrectly.	Inspect for worn or "out of round" cylinder. Install new rings. Break in new rings with engine under varying loads.

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